

Identification of Marginal Treatment Effects using Subjective Expectations

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We develop a nonparametric method that identifies marginal treatment effects (*MTEs*) using survey data on subjective expectations instead of instruments. Subjective treatment probabilities rank people by their latent propensity to be treated if they forecast treatment the same way. Combined with expected treatment-contingent outcomes, the ranking identifies the full *MTE* function if those expectations are correct on average. Both results hold under common forms of reporting error. We apply the method to study childbirth and female labor supply in Denmark. Estimated *MTEs* are roughly constant across ranks of fertility beliefs, so there is little selection into childbirth on unobserved determinants of work after birth. The survey-based average treatment effect on the treated nearly equals the event-study estimate from administrative data at every horizon, evidence that the identifying assumptions hold. Furthermore, eliciting beliefs under a counterfactual policy measures its effect before implementation, accounting for how the policy changes selection into treatment.

JEL Codes: C32, C52, C83, J13, J22

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1 Introduction

Estimating the causal effect of a treatment on outcomes is a central problem in applied economics. The problem is challenging because the effect of treatment varies across individuals and because selection into treatment is driven by unobserved factors that also affect outcomes. Quasi-experimental methods address selection, but their estimates apply to the subpopulation whose treatment the quasi-experiment moves. Whether the estimated effects extend to other subpopulations, their external validity, is often unknown. Marginal treatment effects (*MTEs*) are the parameters that account for both heterogeneity and selection.¹ An *MTE* is the average treatment effect among individuals with the same latent propensity to be treated, and any average treatment effect, such as that on the treated (*ATT*) or on the untreated (*ATU*), is a weighted average of *MTEs*.

This paper develops a nonparametric method that identifies *MTEs* using survey data on subjective expectations. The standard method identifies *MTEs* with instrumental variables, which exogenously shift the latent selection margin, and compares average outcomes around the shift. Nonparametric identification for the full population requires an instrument whose induced propensity score varies across the full support of the latent propensity to be treated. Such instruments are rare, so in practice identification relies on parametric assumptions or delivers only bounds on the treatment effects of interest. Our method uses no instrument. Instead of shifting the selection margin, we locate each individual on the latent rank using elicited treatment beliefs and pair that rank with elicited treatment-contingent outcome expectations to identify the *MTE* directly.

The key insight is that individuals know more than the econometrician about their selection into treatment. The unobserved factors that drive selection include the individual's preferences, costs, and beliefs. The econometrician does not observe these factors in naturally occurring observational data, but the individual's subjective treatment probability depends on them. Selection may also depend on events that have not yet occurred at the time of the survey, so we elicit a treatment probability rather than a deterministic treatment decision.

Our main methodological contribution is to identify the latent propensity to select into treatment and *MTEs* using subjective expectations. First, the subjective treatment probability identifies the latent determinant of selection into treatment if people with the same observable characteristics use a common model to forecast their selection. The identification covers the full support of the latent determinant because every respondent is ranked. Second, combining the identified latent treatment propensity with subjective expectations about

¹See, e.g., Heckman and Vytlacil (1999), Heckman and Vytlacil (2005), Heckman and Vytlacil (2007a), and Eisenhauer, Heckman, and Vytlacil (2015).

treatment-contingent outcomes identifies subjective marginal treatment effects (*seaMTEs*). Third, if in addition individuals' subjective expected gains from treatment are correct, then subjective and objective *MTEs* are equal. Thus, it is sufficient, but not necessary, for individuals to have rational expectations. We also show that the method is robust to common types of error in the measurement of subjective expectations.

The same subjective treatment probabilities, used as weights rather than as ranks, combine with the outcome expectations to identify average treatment effects, such as the *ATT* and *ATU*, under conditions on the levels of beliefs and no rank condition. Furthermore, eliciting beliefs under a counterfactual policy regime identifies its policy relevant treatment effect, with no assumption that the policy leaves treatment-contingent outcomes unchanged.

As a proof of concept, we apply the method with childbirth as the treatment and female labor supply in Denmark as the outcome. Childbirth is a choice women have likely considered carefully and about which they hold private information. We estimate *seaMTEs* using a custom-designed survey that elicits beliefs about the likelihood of childbirth in the near term and birth-contingent labor supply from a representative sample of about 11,000 Danish women aged 18–40. The estimated *MTEs* are roughly constant across levels of fertility beliefs, so after conditioning on typical demographic and economic observables there is little selection into having children based on a latent propensity to work after having a child. Having a child reduces the likelihood of working by about 50 percentage points 9 months after birth, around 3 percentage points 18 months after, and essentially zero 36 months after.

The survey-based estimates match realized behavior in administrative data. Linking respondents to the Danish registries permits two comparisons. First, fertility beliefs have the same conditional correlations with demographic and economic characteristics as realized fertility. Second, the *ATT* computed from our estimated *seaMTEs* is nearly equal, at every elicited horizon from 3 months before birth to 36 months after, to the event-study *ATT* estimated from the realized labor supply of the same women around childbirth. The two *ATTs* are weighted averages of the *seaMTE* and objective *MTE* functions, respectively. Thus, the alignment is evidence that the identifying assumptions hold and that $seaMTE = MTE$ in this context. Passing this check is valuable because the validated *seaMTE* function then delivers every average treatment effect of interest, not only the *ATT* that the event study identifies.

We also use *seaMTEs* to predict the effects of policies that have never been implemented, accounting for how a policy changes selection into treatment. In the survey we elicit childbirth and birth-contingent labor supply probabilities under two counterfactual childcare regimes. Making childcare free raises expected fertility by about 5 percentage points and leaves the labor supply path around a birth and the *MTE* function unchanged, consistent with policy

invariance of the baseline *MTEs*. Tripling the price of childcare and lowering its quality lowers expected fertility by about 10 percentage points and expected labor supply among mothers by about 32 percentage points 36 months after birth. The *MTE* function is shifted down, so baseline *MTEs* are not policy invariant.

1.1 Relationship to Literature

We contribute to the body of research by Heckman and Vytlacil (1999, 2001, 2005, 2007b) that develops and uses *MTEs* to study unobserved heterogeneity and the external validity of treatment effect estimates. Heckman and Vytlacil (1999) shows that *MTEs* can be estimated by the method of local instrumental variables. However, nonparametric identification of *MTEs* for the full population requires an instrument powerful enough to move the propensity score continuously across its full support, conditional on all combinations of conditioning variables. This requirement is stringent, and in practice extra assumptions are invoked to gain identification.

One approach is to use point-identified treatment effects and either assume no treatment effect heterogeneity (see, e.g., Heckman and Robb (1985), Angrist and Fernandez-Val (2013)) or parameterize unobserved heterogeneity. Prominent examples include assuming normality of the idiosyncratic selection and outcome components (Bjorklund and Moffitt (1987), Heckman, Tobias, and Vytlacil (2003)), full independence assumptions under which the *MTE* slope does not depend on the conditioning variables X (see, e.g., Aakvik, Heckman, and Vytlacil (2005); Carneiro, Heckman, and Vytlacil (2011); Carneiro, Lokshin, and Umapathi (2017)), and additive separability (Brinch, Mogstad, and Wiswall (2017)). A second approach bounds treatment effect parameters using what observational data identify (see, e.g., Manski (1990), Manski (1997), Manski and Pepper (2000), Manski (2003), Heckman and Vytlacil (2007b)). A recent advance is Mogstad, Santos, and Torgovitsky (2018), which uses the full set of IV-like estimands identified in the data to constrain marginal treatment response functions and infer nonparametric bounds on *MTEs* that are not point-identified.

Our method differs from this previous work in that it nonparametrically point-identifies *seaMTEs* over the entire support of the latent determinant of selection into treatment using subjective expectations data and no instrument, and identifies the objective *MTEs* under the outcome-belief conditions we state. This approach allows us to study the external validity of local average treatment effects (*LATEs*) and other treatment effect parameters and to identify treatment effects for subpopulations that lack sufficient observational data on treatment status, outcomes, and instruments. Section 2.6 discusses in detail how the survey-based approach relates to the instrumental variables approach.

Recently, de Paula and Tamer (2026) also connect instruments to elicited probabilities. They show that an instrument identifies an average of individual treatment effects weighted by how much it moves each individual’s treatment probability, and they point to elicited probabilities as a route to the full *MTE* function. We supply the measurement and the identifying conditions for that route.

Our paper also relates to the literature on subjective expectations. Following Manski (2004)’s survey of early work in this area, a number of studies measure subjective expectations and show that they matter for economic choices and outcomes.² We relate most closely to papers that elicit state-contingent outcomes. A causal treatment effect contrasts treatment-contingent outcomes for the same person, holding fixed everything else that affects outcomes. Data on realized outcomes cannot deliver this within-person comparison, which is the fundamental evaluation problem (Rubin, 1974). Eliciting subjective state-contingent outcomes addresses this problem directly.

Asking about state-contingent expectations dates back to at least Dominitz and Manski (1996), who, in an exploratory study, asked high school students and college undergraduates about probabilistic expectations under alternative scenarios for future schooling. A number of recent studies use subjective expectations of state-contingent outcomes to model educational choices.³ Closely related to our study, Arcidiacono, Hotz, Maurel, and Romano (2020) study occupational choice using data for 173 Duke undergraduates on occupation-specific earnings beliefs, the probabilities of choosing each occupation, and realizations. They show how these data recover ex ante individual treatment effects of occupational choice on earnings and how the individual effects relate to the ex ante average treatment effect and the ex ante average treatment effect on the treated. A similar approach is applied by Giustinelli and Shapiro (2024), who collect beliefs about health status and retirement and use them to estimate individual treatment effects of health on labor supply. See also Hudomiet, Hurd, Parker, and Rohwedder (2021), who use the subjective conditional probability of working at age 70 to estimate the causal effect of job characteristics on retirement.

Relative to these studies, we show how and under what assumptions subjective expectations

²See, e.g., Pistaferri (2001), Lochner (2007), Delavande (2008), Attanasio (2009), Hurd (2009), Delavande and Rohwedder (2011), van der Klaauw (2012), Delavande (2014), Delavande and Kohler (2016), Hendren (2017), Giustinelli and Manski (2018), Alesina, Stantcheva, and Teso (2018), Bailey, Davila, Kuchler, and Stroebel (2019), Kuchler and Zafar (2019), Christelis, Georgarakos, Jappelli, and van Rooij (2020), Coibion, Gorodnichenko, and Ropele (2019), Wiswall and Zafar (2021), Andersen and Leth-Petersen (2021), and Colarieti, Mei, and Stantcheva (2024). Also using survey expectations, Jappelli and Pistaferri (2014) and Fuster, Kaplan, and Zafar (2021) investigate the effect of unobserved state-contingent choices in the context of spending responses to fiscal policy.

³See Arcidiacono, Hotz, and Kang (2012), Zafar (2011), Zafar (2013), Stinebrickner and Stinebrickner (2013), Wiswall and Zafar (2015), Kaufmann (2014), Attanasio and Kaufmann (2014), Attanasio and Kaufmann (2017), and Delavande and Zafar (2019).

of the treatment propensity and of state-contingent outcomes identify *MTEs* and, by extension, any conventional average treatment effect parameter. That is, in addition to the distribution of individual treatment effects that these studies measure, we address nonrandom selection into treatment. We also specify the conditions under which historical administrative data validate treatment effects estimated from subjective expectations, an alternative to waiting until choices and outcomes are realized, as Arcidiacono, Hotz, Maurel, and Romano (2020) and Wiswall and Zafar (2021) do. Finally, we provide a method to infer treatment effects in counterfactual policy environments, which yields a suggestive test of the policy invariance of *MTEs* estimated from historical data.

We are unaware of another study that documents the role of unobserved heterogeneity in the effect of childbirth on labor supply. Within this large literature, two recent papers study female labor supply after childbirth in Denmark. Kleven, Landaais, and Sogaard (2019) use an event study design and administrative data on first births during 1985–2003 and find that childbirth reduces female labor supply by about 12–14 percent. Lundborg, Plug, and Rasmussen (2017) exploit randomness in the success of in vitro fertilization (IVF) treatments to construct IV estimates for women with no prior children. They find that IVF-treated women who have children reduce labor force participation by about 7 percentage points in the two years after birth compared to IVF-treated women who do not. Although IVF-treated women are a selected group, these estimates are of similar magnitude to those of Kleven, Landaais, and Sogaard (2019). We show that selection on unobserved characteristics is unlikely to be important in the Danish context, which ties together the findings of Kleven, Landaais, and Sogaard (2019) and Lundborg, Plug, and Rasmussen (2017).

2 A Survey Method to Identify Marginal Treatment Effects

We take the Generalized Roy model (Roy, 1951) as the data generating process and show how subjective expectations identify *MTEs*. To ease exposition, we model scalar shocks and specify a structural outcome function. Appendix B extends all results to multi-dimensional shocks, restates the environment as a general potential-outcomes model that nests this one, and extends the results to treatment effects on general functionals of the outcome distributions (e.g., distributional treatment effects). We present the model abstractly and then interpret each variable in the context of female labor supply and childbirth.

2.1 The Model

Let $D^i \in \{0, 1\}$ indicate whether individual i is treated. Let $Y_D^i \in \mathbb{R}$ be the potential outcome for individual i when treatment status is set to D . Selection into treatment depends on the difference between a random variable μ^i and an idiosyncratic component $V^i \in \mathbb{R}$:

$$D^i = \begin{cases} 1 & \text{if } \mu^i \geq V^i \\ 0 & \text{if } \mu^i < V^i. \end{cases} \quad (1)$$

V^i captures unobserved preferences and costs, which we call the distaste for treatment.

The model has three dates, matched to the timing of the survey and of events. By $t = 0$, the time of the survey, the individual constants $X^i \in \mathbb{R}^{d_x}$ and V^i are determined. At $t = 1$, the random variable μ^i is drawn from the CDF $F_{\mu|X^i}$, which depends on X^i , and treatment status D^i is determined by selection rule (1). At $t = 2$, the random variable ϵ^i is drawn from the CDF $F_{\epsilon|X^i}$, with $\mu^i \perp \epsilon^i | X^i$ and $(\mu^i, \epsilon^i) \perp V^i | X^i$. The outcome Y_D^i is determined in period $t = 2$. Crucially, the outcome is modeled as a function of treatment status (D^i), individual characteristics (X^i, V^i) and shocks (μ^i, ϵ^i): $Y_D^i(X^i, V^i, \mu^i, \epsilon^i)$. The function is defined at every realization of (μ^i, ϵ^i) , including values of μ^i at which selection rule (1) would assign the other treatment state, so Y_1^i is the outcome with treatment whether or not treatment would have been chosen.

Selection rule (1) generalizes the standard selection framework (see, e.g., Heckman and Vytlacil (2005)). Both V and X are individual characteristics that affect selection into treatment. As usual, the key difference is that X is observable to the econometrician, while V is not. The generalization is that the first term in the selection rule, μ , is stochastic. Because individuals are often surveyed before selection into treatment occurs and cannot perfectly forecast their own selection, the stochastic case is relevant for many applications.

Information, Beliefs, and Behavior. At $t = 0$, before selection into treatment occurs, the individual knows X^i, V^i , but does not know μ^i or ϵ^i (and thus does not know D^i or Y_D^i). Individual i 's beliefs at $t = 0$ are a subjective probability of treatment, $G^i \in [0, 1]$, and, for each treatment state D , a subjective probability density h_D^i over the potential outcome Y_D^i .⁴ These beliefs are primitives. No relationship between G^i and h_D^i is imposed, and neither need be correct. If the individual has rational expectations, i.e., if their subjective beliefs align

⁴For discrete outcomes—as in our application, where the outcome is binary— h_D^i is a probability mass function and integrals over y denote sums. Appendix B treats general outcome spaces formally.

with the probabilities dictated by the model, then

$$G^i = 1 - F_{\mu|X^i}(V^i). \quad (2)$$

Let $\mathbb{E}^i[Y_D^i]$ be the treatment-contingent subjective expected value of the outcome, the mean of h_D^i over outcome values y , and let $seaITE^i$ be the subjective ex ante individual treatment effect, i.e., the subjective expected difference in outcomes across treatment states:

$$\mathbb{E}^i[Y_D^i] := \int y h_D^i(y) dy, \quad (3)$$

$$seaITE^i := \mathbb{E}^i[Y_1^i] - \mathbb{E}^i[Y_0^i]. \quad (4)$$

The treatment state enters h_D^i only through its subscript. The density h_D^i describes the individual's beliefs about Y_D^i with the treatment state fixed at D , so $\mathbb{E}^i[Y_D^i]$ averages over their uncertainty about μ^i and ϵ^i and does not condition on the event $D^i = D$. If the individual has rational expectations, h_D^i is the density of $Y_D^i(X^i, V^i, \mu^i, \epsilon^i)$ implied by $F_{\mu|X^i}$ and $F_{\epsilon|X^i}$, so equation (3) equals the expectation of the potential-outcome function over the shocks,

$$\mathbb{E}^i[Y_D^i] = \int \int Y_D^i(X^i, V^i, \mu, \epsilon) dF_{\epsilon|X^i}(\epsilon) dF_{\mu|X^i}(\mu). \quad (5)$$

Individuals' preferences and subjective beliefs affect their behavior. The model captures this by allowing the V^i that determines selection into treatment to depend on the subjective outcome beliefs $h_D^i(\cdot)$ as well as on unspecified idiosyncratic components such as preferences and costs. Because state-contingent outcomes depend on V^i , any statistic built from differences in state-contingent outcomes, such as the MTE , also depends on these beliefs.

Selection into treatment is determined by V^i and the realization of μ^i , not by the treatment belief G^i . The treatment belief affects what individuals report in the survey at $t = 0$, not treatment status or outcomes.

Example: Female Labor Supply and Childbirth through the Lens of the Model. To make this concrete, consider a woman who may give birth to a child 24 months in the future and may be working 36 months in the future. The treatment is having the child and the outcome is working. Characteristics in our data such as age, income, wealth, and partner status (X^i) may affect her propensity to have a child, which the model captures by letting the distribution of μ^i depend on X^i . Whether she loves being around children and whether she has access to low-cost childcare may affect her desire to have a child but are not measured in our data. Such preferences and costs are captured by V^i . Her beliefs about how childbirth would affect

her employment ($h_D^i(\cdot)$) can also influence her decision to have a child, and these too are captured by V^i . She may have a chance at a promotion, and whether she gets it may affect her desire to have a child, which is captured by the stochastic component μ^i . Her desire to work, with or without a child, may also depend on whether she got the promotion, which is why the outcome function $Y_D^i(X^i, V^i, \mu^i, \epsilon^i)$ depends directly on μ^i and not only on D . Her desire to work after having a child may depend on whether she loves being around children, which is why the outcome function depends directly on V^i . Finally, an event that affects her desire to work independently of child status and promotion status, such as a parental illness, is captured by ϵ^i .

Normalization and Notation. As is common in the literature, we assume V is continuous.⁵ Similarly, we assume G is continuous.

Assumption 1 (Continuity). *V and G are continuously distributed, conditional on X .*

This assumption allows us to normalize the latent selection component to the unit interval. Define the CDF of V conditional on X as $F_{V|X}$ and let

$$U^i := F_{V|X^i}(V^i). \quad (6)$$

U^i denotes individual i 's order in the distribution of latent selection component V , conditional on X .⁶

Similarly, define the CDF of G conditional on X as $F_{G|X}$ and let $\tilde{U}^i := 1 - F_{G|X^i}(G^i)$. Then $\tilde{U}^i$ is individual i 's order in the distribution of the subjective probability of selecting into treatment, conditional on X , when ordered from highest to lowest.

2.2 Definition of MTE s

Heckman and Vytlacil (1999) and Heckman and Vytlacil (2005) define the MTE as the expected gain from treatment ($Y_1^i - Y_0^i$) conditional on X and U .

Definition of Marginal Treatment Effects (MTE s).

$$\begin{aligned} MTE(u, x) &:= \mathbb{E} [Y_1^i - Y_0^i | U^i = u, X^i = x] \\ &= \int \int [Y_1^i(x, v_u, \mu, \epsilon) - Y_0^i(x, v_u, \mu, \epsilon)] dF_{\epsilon|x}(\epsilon) dF_{\mu|x}(\mu), \end{aligned} \quad (7)$$

⁵See, e.g., Assumption 3 in Vytlacil (2002).

⁶We adopt almost-sure conventions without further comment. Equation (2) holds for almost every individual because a CDF has at most countably many discontinuity points and V is atomless. Conditional on $U^i = u$ and $X^i = x$, V^i equals the u -quantile of $F_{V|x}$ for almost every u .

where $v_u := F_{V|x}^{-1}(u)$ is the value of the latent component corresponding to rank u . Conditioning on $(U^i = u, X^i = x)$ fixes the type at $(V, X) = (v_u, x)$, so the second line averages only over the shocks (μ, ϵ) , whose distributions given X do not depend on V . Like the subjective expectation in equation (3), it averages over all values of μ , here under the model's distribution rather than the individual's. Under rational expectations the two distributions coincide, and the integral is the difference of the two integrals in equation (5) evaluated at (x, v_u) .

The *MTE* is the limit form of the *LATE* (Heckman and Vytlačil, 1999). It is an average treatment effect that is local to individuals with the same X^i and the same latent propensity to select into treatment U^i . It is an ex ante object because the expectation is taken over the shocks that determine selection and outcomes, under the model's distributions, conditional on the information available before either is realized.

Definition of Subjective Ex Ante *MTEs* (*seaMTEs*). The subjective ex ante marginal treatment effect (*seaMTE*) is the same object with both objective ingredients replaced by their subjective counterparts. It conditions on the subjective treatment propensity, $\tilde{U}^i$ instead of U^i , and it takes expectations under the individual's subjective distributions, $\mathbb{E}^i[\cdot]$ instead of $\mathbb{E}[\cdot]$:

$$\begin{aligned} seaMTE(u, x) &:= \mathbb{E} \left[seaITE^i | \tilde{U}^i = u, X^i = x \right] \\ &= \mathbb{E} \left[\int y [h_1^i(y) - h_0^i(y)] dy \mid \tilde{U}^i = u, X^i = x \right]. \end{aligned} \tag{8}$$

Unlike in the *MTE*, the outer expectation in the *seaMTE* is not degenerate. Individuals who share the same $(\tilde{U}, X)$ may still hold different subjective outcome beliefs h_D^i , because $\tilde{U}$ is a function of the treatment belief alone, and the *seaMTE* averages over this residual belief heterogeneity.

If individuals have rational expectations, then $seaMTE(u, x) = MTE(u, x)$. The two objects can differ for two reasons. First, individuals may not have rational expectations about μ . In the running example, a woman who believes she is more likely to get the promotion than she is has a subjective probability of having a child that differs from the actual chance. Her rank in the distribution of subjective treatment probabilities ($\tilde{U}^i$) may then differ from her rank in the distribution of the latent distaste for treatment (U^i), so the two objects condition on different groups of individuals. Second, individuals may not have rational expectations about outcomes. For example, Kuziemko, Pan, Shen, and Washington (2018) document a “Mommy Effect” in which women think that they are more likely to work after childbirth than they really are. The model allows for this if, e.g., ϵ captures how well the child tolerates being away from the mother, mothers are overoptimistic about this, and mothers are less

likely to work if their child does not tolerate being away from them.

Remark: Ex Ante and Ex Post. The *MTE* in equation (7) is an ex ante object. Conditional on (U^i, X^i) , the gain $Y_1^i - Y_0^i$ is a random variable whose distribution is induced by the shocks (μ, ϵ) , and the *MTE* is its mean. It is useful to distinguish two ex post objects. The first is the average realized gain among individuals with $U^i = u$ and $X^i = x$. By the law of large numbers this average equals $MTE(u, x)$. Realizations average back to the ex ante mean, so in this sense nothing separates ex ante from ex post. The second is the ex post treated–untreated difference, which compares realized outcomes of treated and untreated individuals within a (U, X) cell,

$$\mathbb{E} [Y^i | D^i = 1, U^i = u, X^i = x] - \mathbb{E} [Y^i | D^i = 0, U^i = u, X^i = x], \quad (9)$$

where $Y^i := Y_0^i(1 - D^i) + Y_1^i D^i$ is the realized outcome. Within the cell, treatment status is determined entirely by the realization of μ^i (since $D^i = 1$ if and only if $\mu^i \geq v_u$). Because outcomes may depend directly on μ^i , treated and untreated members of the same cell differ systematically in μ^i : the first term in (9) averages Y_1 over high realizations of μ and the second averages Y_0 over low realizations. The difference (9) therefore does not equal $MTE(u, x)$ in general. Thus conditioning on the latent type does not remove the classic selection problem (Heckman and Robb, 1985), because within a type μ determines both treatment and outcomes (Cunha, Heckman, and Navarro, 2005). The two coincide when outcomes do not depend on μ , i.e., when the outcome function takes the form $Y_D^i(X^i, V^i, \epsilon^i)$. In the running example, women in a given cell who have a child are disproportionately those who got the promotion, and the promotion directly affects their propensity to work, which contaminates the within-cell comparison of mothers and non-mothers.

Our survey approach targets the ex ante *MTE*. Each respondent’s expectations integrate over her own uncertainty about μ (equation (3)), so no selection on μ enters. This is also why the survey contains information that a single cross-section of realized choices does not. de Paula and Tamer (2026) show that an ex ante stochastic choice model and an ex post deterministic one generate identical first stages, reduced forms, and Wald estimands. Realized choices alone therefore cannot distinguish ex ante uncertainty from fixed heterogeneity, and so cannot recover the ex ante choice probability. Elicitation measures it directly.

2.3 Identifying the Latent Treatment Propensity and the *MTE*

The econometrician observes X but not V , and has survey data, collected at $t = 0$ before selection into treatment, on each individual’s reported subjective probability of selecting into

treatment, $\hat{G}^i$, and reported state-contingent outcome distributions, $\hat{h}_D^i$. Both G^i and h_D^i are subjective probability distributions, so surveys can measure them directly.

A survey response can differ from the corresponding objective probability for two reasons. First, the individual's belief may be wrong, in that it differs from the probability the data generating process implies. Second, the belief may be right but measured with error. This section allows the first and rules out the second. We assume the survey measures beliefs exactly, $\hat{G}^i = G^i$ and $\hat{h}_D^i = h_D^i$, so the results identify subjective objects from correctly measured beliefs. Section 2.5 relaxes the measurement assumption.

Identifying the Latent Treatment Propensity U^i . Assumption 2 provides the condition under which U^i is identified from subjective expectations data.

Assumption 2 (Common Forecasting Rule). *For each x in the support of X there exists a CDF $\tilde{F}_{\mu|x}$, common to all individuals with $X^i = x$, such that $G^i = 1 - \tilde{F}_{\mu|x}(V^i)$.*

Under Assumption 2, individuals with the same X^i differ in their subjective probability of selecting into treatment only because they differ in V^i , and those with higher V^i have a lower subjective probability. The CDF $\tilde{F}_{\mu|X^i}$ is a forecasting rule that maps X^i and V^i into a probability of selecting into treatment, and the assumption requires that people with the same X^i use the same rule. Rational expectations, $\tilde{F}_{\mu|X^i} = F_{\mu|X^i}$, satisfy the assumption, but only the commonality of the rule is needed, not its correctness.

Analogous to U^i and $\tilde{U}^i$, define $\hat{U}^i := 1 - F_{\hat{G}|X^i}(\hat{G}^i)$ to be the order of individuals with the same observable characteristics when ranked from highest to lowest by their reported subjective probability of selecting into treatment.

Proposition 1 (Subjective Treatment Probabilities Identify Latent Treatment Propensities). *Let Assumptions 1, 2, and zero measurement error hold. Then*

$$\hat{U}^i = \tilde{U}^i = U^i.$$

Proof: See Appendix A.

Proposition 1 is a main result of the paper because it relates a measured object to the latent parameter that determines selection into treatment. With zero measurement error, the reported rank is the subjective rank. Under Assumption 2, measuring the subjective probability of selection into treatment makes the latent treatment propensity U^i observed, so *MTEs* can be identified by conditioning on it directly.

The idea is simple. Consider two individuals, i and j , who have the same observable characteristics X . Given Assumption 2, they use the same forecasting rule, so they differ

in their subjective probability of being treated only through V^i and V^j . The probability of receiving treatment is decreasing in V and V is in an individual's information set, so individual i has a higher subjective probability of selection into treatment than individual j if and only if $V^i < V^j$.⁷

Identifying the Subjective MTE ($seaMTE$). The individual treatment effect (ITE) is the difference between an individual's treatment-contingent outcomes. It cannot be measured from realized outcomes because no individual is observed both in and out of treatment. Its subjective ex ante counterpart, the $seaITE$ of equation (4), can be identified from a single cross-section of survey data on subjective expectations of outcomes. Prior work uses subjective expectations about outcomes to estimate ITE s and average treatment effects.⁸ Our contribution is to address nonrandom selection into treatment and identify the MTE by also identifying U^i .

Let $\widehat{seaITE}^i$ be the $seaITE$ computed using the directly elicited state-contingent probability of outcomes from the survey:

$$\widehat{seaITE}^i := \widehat{\mathbb{E}}^i[Y_1^i] - \widehat{\mathbb{E}}^i[Y_0^i] = \int y [\hat{h}_1^i(y) - \hat{h}_0^i(y)] dy. \quad (10)$$

Similarly, let $\widehat{seaMTE}$ be the $seaMTE$ computed using the elicited $\widehat{seaITE}$ s and subjective treatment propensities:

$$\widehat{seaMTE}(u, x) := \mathbb{E} \left[\widehat{seaITE}^i | X^i = x, \hat{U}^i = u \right]. \quad (11)$$

With zero measurement error, $\hat{U}^i = \tilde{U}^i$ and $\widehat{seaITE}^i = seaITE^i$, so the measured $seaMTE$ is the $seaMTE$, $\widehat{seaMTE}(u, x) = seaMTE(u, x)$, and the $seaMTE$ function is identified from individual characteristics and reported beliefs alone.

When do Subjective MTE s Equal Objective MTE s? The $seaMTE$ is a subjective object. Whether it equals the objective MTE depends on whether beliefs are correct. Full rational expectations are not required: it suffices that individuals rank themselves correctly (Proposition 1) and that their expected gain from treatment is correct (Assumption 3).

⁷Imbens and Newey (2009) recover the rank of a scalar unobservable from the conditional distribution of an endogenous regressor given instruments and use it as a control variable for realized outcomes. Our rank argument is analogous, with the treatment probability as regressor and the covariates as instruments. We do not, however, use U^i as a control variable for realized outcomes, because within a cell the selection shock still moves both treatment and outcomes (equation (9)).

⁸See, e.g., Arcidiacono, Hotz, Maurel, and Romano (2020), Wiswall and Zafar (2021), and Giustinelli and Shapiro (2024).

Assumption 3 (Mean-Correct Gain Beliefs).

$$\mathbb{E}^i [Y_1^i] - \mathbb{E}^i [Y_0^i] = \mathbb{E} [Y_1^i - Y_0^i | X^i, V^i] \quad \text{for all } i.$$

Proposition 2 (Mean-Correct Beliefs Equate *seaMTE* and *MTE*). *Let Assumptions 1, 2, 3, and zero measurement error hold. Then*

$$\widehat{seaMTE}(u, x) = seaMTE(u, x) = MTE(u, x).$$

Proof: See Appendix A.

With zero measurement error, $\widehat{seaMTE} = seaMTE$. The equality $seaMTE = MTE$ clarifies exactly how correct beliefs must be. Individuals may misperceive their treatment probability G^i in any way consistent with the common forecasting rule (Assumption 2). They may also hold incorrect beliefs about any feature of the outcome distributions other than the difference in means, so expected outcomes may be wrong in both treatment states by a common amount (Assumption 3). All that matters is that the ranking of V is preserved and that the subjective expected gain from treatment is right.⁹ Rational expectations imply Assumption 2 (with $\tilde{F}_{\mu|x} = F_{\mu|x}$) and Assumption 3 (equation (5)) and are therefore sufficient, but not necessary.

2.4 Average Treatment Effects and Policy Counterfactuals

Average Treatment Effects. Identifying the *MTE* function uses the rank conditions, Assumptions 1 and 2, because the *MTE* conditions on the latent rank. Average treatment effects, including the unconditional average treatment effect (*ATE*), do not condition on the rank, so they are identified under conditions on the *levels* of beliefs and no rank condition.

Assumption 4 (Mean-Correct Treatment Beliefs).

$$\mathbb{E} [G^i | X^i = x, V^i = v] = P(D^i = 1 | X^i = x, V^i = v) \quad \text{for each } (x, v) \text{ in the support of } (X, V).$$

Proposition 3 (Average Effects from Belief Levels). *Let zero measurement error and Assumption 3 hold. Then:*

$$(i) \quad \mathbb{E} [\widehat{seaITE}^i] = \mathbb{E} [Y_1^i - Y_0^i] = ATE;$$

⁹For Proposition 2, Assumption 3 can be weakened to hold on average within a type, $\mathbb{E}[seaITE^i | X^i = x, V^i = v] = \mathbb{E}[Y_1^i - Y_0^i | X^i = x, V^i = v]$ for each (x, v) in the support of (X, V) .

(ii) In addition, let Assumption 4 hold and $\mathbb{E}[G^i] \in (0, 1)$. Then

$$\begin{aligned} ATT &:= \frac{\mathbb{E}[\widehat{seaITE}^i \times \hat{G}^i]}{\mathbb{E}[\hat{G}^i]} = \mathbb{E} \left[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] \mid D^i = 1 \right], \\ ATU &:= \frac{\mathbb{E}[\widehat{seaITE}^i \times (1 - \hat{G}^i)]}{\mathbb{E}[1 - \hat{G}^i]} = \mathbb{E} \left[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] \mid D^i = 0 \right]. \end{aligned}$$

Proof: See Appendix A.

Proposition 3 does not use Assumption 1 or 2. Part (i) requires nothing about treatment beliefs. Part (ii) uses the levels of $\hat{G}$, not its ranks. If the rank conditions fail, the *MTE* function is not identified, but the *ATE*, the *ATT*, and other average treatment effects are.

Policy-Relevant Treatment Effects. The same logic extends to policy counterfactuals. In the framework of Heckman and Vytlacil (2005), the policy relevant treatment effect (*PRTE*) of moving from a baseline policy to an alternative is constructed by re-weighting baseline *MTEs*. This construction requires policy invariance: the policy moves the treatment margin and leaves the expected outcome in each treatment state, given the latent type, unchanged. The *MTE* function is then unchanged as well. Eliciting beliefs directly in counterfactual scenarios removes this requirement.

For policy regime $p \in \{b, c\}$ (baseline, counterfactual), let D_p denote treatment and $Y_{D,p}$ the potential outcomes under regime p , with realized outcome $Y_p = (1 - D_p)Y_{0,p} + D_pY_{1,p}$. Let G_p^i be the elicited probability of treatment under regime p and $\mathbb{E}_p^i[Y_{D,p}^i]$ the elicited expected outcome in treatment state D under regime p . Assumptions 3 and 4 in regime p refer to these objects and to D_p and $Y_{D,p}$. The ex ante regime mean weights, for each individual, the expected outcome in each treatment state by the probability of that state. Its subjective version, $\bar{Y}_p^{sea}$, uses each respondent's own treatment probability and treatment-contingent outcome expectations, averaged across respondents,

$$\bar{Y}_p^{sea} := \mathbb{E} [G_p^i \mathbb{E}_p^i[Y_{1,p}^i] + (1 - G_p^i) \mathbb{E}_p^i[Y_{0,p}^i]]. \quad (12)$$

Its objective version, $\bar{Y}_p$, uses each individual's objective treatment probability and objective expected outcomes given her type,

$$\bar{Y}_p := \mathbb{E} [P(D_p = 1 | X, V) \mathbb{E}[Y_{1,p} | X, V] + P(D_p = 0 | X, V) \mathbb{E}[Y_{0,p} | X, V]]. \quad (13)$$

The subjective ex ante $PRTE$ is the change in the subjective regime mean per unit change in the expected treatment share, and the ex ante $PRTE$ is its objective counterpart,

$$seaPRTE := \frac{\bar{Y}_c^{sea} - \bar{Y}_b^{sea}}{\mathbb{E}[G_c] - \mathbb{E}[G_b]} \quad \text{and} \quad PRTE := \frac{\bar{Y}_c - \bar{Y}_b}{\mathbb{E}[D_c] - \mathbb{E}[D_b]}, \quad (14)$$

when the denominators are nonzero. The measured $\widehat{seaPRTE}$ replaces G_p^i and $\mathbb{E}_p^i[\cdot]$ in equation (12) by their reports, giving $\hat{Y}_p^{sea}$ in place of $\bar{Y}_p^{sea}$ and $\mathbb{E}[\hat{G}_p]$ in place of $\mathbb{E}[G_p]$. The $PRTE$ is an ex ante object, like the MTE .¹⁰

When potential outcomes themselves change across regimes, the ex ante $PRTE$ bundles the effect of treatment with the direct effect of the policy on treatment-contingent expected outcomes, rather than reweighting baseline MTE s. Because the survey elicits treatment-contingent expectations under each regime, policy invariance becomes a testable hypothesis rather than an identifying assumption. Comparing the expectations elicited under the baseline and counterfactual regimes tests it directly, and comparing the MTE functions tests its implication for the treatment gain.

Proposition 4 (Identification of the $PRTE$ without Policy Invariance). *Let zero measurement error hold in each policy regime $p \in \{b, c\}$. Then:*

- (i) $\widehat{seaPRTE} = seaPRTE$;
- (ii) if Assumptions 3 and 4 hold in each regime and the average bias in the level of outcome beliefs is the same in both regimes, $\mathbb{E}[\mathbb{E}_c^i[Y_{0,c}^i] - \mathbb{E}[Y_{0,c}^i | X^i, V^i]] = \mathbb{E}[\mathbb{E}_b^i[Y_{0,b}^i] - \mathbb{E}[Y_{0,b}^i | X^i, V^i]]$, then $\mathbb{E}[G_p^i] = \mathbb{E}[D_p]$ and $seaPRTE = PRTE$;
- (iii) if, in addition, treatment is mean-independent of each potential outcome within type, $\mathbb{E}[Y_{D,p} | D_p, X, V] = \mathbb{E}[Y_{D,p} | X, V]$ for $D \in \{0, 1\}$, then $\bar{Y}_p = \mathbb{E}[Y_p]$, and the $PRTE$ equals the change in mean realized outcomes per unit change in the treatment share, $\frac{\mathbb{E}[Y_c] - \mathbb{E}[Y_b]}{\mathbb{E}[D_c] - \mathbb{E}[D_b]}$.

Proof: See Appendix A.

Part (i) identifies the $seaPRTE$ from elicited beliefs alone, with no restriction linking the two regimes. No policy invariance is imposed. Potential outcomes, selection, and beliefs may all change arbitrarily across regimes, because each regime's objects are elicited within that regime. Part (ii) states when the subjective object equals the $PRTE$. Because Assumption 3 restricts only the expected gain, the level of outcome beliefs may be biased, and the bias

¹⁰Heckman and Vytlacil (2005) define the $PRTE$ on realized outcomes, as $\frac{\mathbb{E}[Y_c] - \mathbb{E}[Y_b]}{\mathbb{E}[D_c] - \mathbb{E}[D_b]}$. In their model the policy shifter is excluded from outcomes and the two definitions coincide. They separate only when the selection shock affects outcomes directly.

must be the same in both regimes so that it cancels. If expected outcomes are correct in each treatment state, not only their difference, there is no bias in either regime, the subjective regime mean equals its objective counterpart, $\bar{Y}_p^{sea} = \bar{Y}_p$, and the survey identifies the level of the outcome under each policy, not only its change. Part (iii) states when the *PRTE* also describes realized outcomes. It holds when, within type, treatment status carries no information about either potential outcome, which in the structural model is the case when outcomes do not depend on the selection shock μ , the same condition under which the ex post treated–untreated difference recovers the *MTE* (Section 2.2).

2.5 Measurement (Survey Response) Error

Sections 2.3 and 2.4 took beliefs as given and assumed the survey measures them exactly. This section again takes beliefs as given, right or wrong, and lets the survey measure them with error. All previous propositions and results hold with Assumptions 5, 6, and 7 below in place of zero measurement error (Appendix A). Table 1 collects, for each parameter, the reporting and belief conditions under which its measured subjective version equals its objective counterpart.

Error in Outcome Beliefs. The *seaMTE*, the average effects, and the *PRTE* are identified even if reported outcome beliefs have a respondent-specific bias common to both treatment states plus a state-specific error that is mean zero given observables and the treatment belief. That is, $\widehat{seaMTE} = seaMTE$ and Propositions 3 and 4 hold with Assumption 5 in place of zero error in outcome reports.

Assumption 5 (Conditionally Unbiased Outcome Reports). *For all i , for $D \in \{0, 1\}$, and for all y , reported beliefs satisfy*

$$\begin{aligned}\hat{h}_D^i(y) &= h_D^i(y) + C^i(y) + \eta_D^i(y), \\ \mathbb{E}[\eta_D^i(y) | X^i, G^i] &= 0.\end{aligned}$$

The first line decomposes the reported density into the true density, a reporting error $C^i(y)$ common to both treatment states, and a state-specific reporting error $\eta_D^i(y)$. The second line requires the state-specific error to be mean zero conditional on observables and the subjective treatment probability. The reported outcome expectation then decomposes as $\widehat{\mathbb{E}}^i[Y_D^i] = \mathbb{E}^i[Y_D^i] + \bar{C}^i + \bar{\eta}_D^i$, where $\bar{C}^i := \int y C^i(y) dy$ is common to both treatment states and $\bar{\eta}_D^i := \int y \eta_D^i(y) dy$ is mean zero conditional on (X^i, G^i) . The common error $\bar{C}^i$ cancels in the $\widehat{seaITE}^i$. The state-specific error $\bar{\eta}_D^i$ makes $\widehat{seaITE}^i \neq seaITE^i$ for individuals, but averaging

the conditionally mean-zero errors within an $(\hat{U}, X)$ cell still gives $\widehat{seaMTE} = seaMTE$. The same two steps deliver the average effects, because an average of $\widehat{seaITE}^i$ weighted by the treatment probability is unchanged by errors that are mean zero given that probability. For the $P RTE$, C^i must also be the same in both policy regimes, so that it cancels across regimes as it cancels across treatment states.

Assumption 5 fails, for example, if the women who overstate their probability of having a child also systematically overstate their probability of working conditional on having a child, since their reporting errors are then correlated with $\hat{U}^i$. The comparison of survey-based and event-study ATT s in Section 3.4.2 tests this condition.

Rank-Preserving Error in Treatment Probabilities. The latent rank and the $seaMTE$ are identified even if reported treatment probabilities are biased, as long as the bias preserves the order of the true probabilities. Both depend on the reports only through the rank, so Proposition 1 holds with Assumption 6 in place of zero measurement error. The equality $\widehat{seaMTE} = seaMTE$ holds with Assumptions 5 and 6, because the reported rank $\hat{U}^i$ then equals the rank $\tilde{U}^i$ in true subjective probabilities on which the $seaMTE$ conditions.

Assumption 6 (Rank-Preserving Treatment Probability Reports). *Suppose that $\hat{G}^i$ is reported such that the order of individuals equals their order in the true subjective probabilities. That is, $G^i < G^j \iff \hat{G}^i < \hat{G}^j \quad \forall i, j$.*

Under the assumption, $\tilde{U}^i = \hat{U}^i$, because the ranks depend only on the order of the reports. For example, if every woman overstates her probability of having a child by 10 percentage points, or if all reports are compressed toward one half by some factor, the levels are wrong but the order is preserved, so U^i is still identified. If reports are only weakly increasing in G , as when respondents round, U^i is set identified within the group of individuals who report the same probability (Corollary A2 in Appendix A).

Binned Ranks. The application estimates MTE s on a coarse partition of the measured rank (Section 3), which requires far less than individual-level rank correctness.

Corollary 1 (Binned MTE s). *Fix a partition $0 = b_0 < b_1 < \dots < b_K = 1$, and let each individual's reported rank $\hat{U}^i$ fall in the same cell of the partition as their latent rank U^i ,*

$$P\left(\mathbb{1}\{U^i \in (b_{k-1}, b_k]\} = \mathbb{1}\{\hat{U}^i \in (b_{k-1}, b_k]\} \text{ for all } k \mid X^i\right) = 1.$$

Then, for every k ,

$$\mathbb{E}\left[seaITE^i \mid \hat{U}^i \in (b_{k-1}, b_k], X^i = x\right] = \mathbb{E}\left[seaITE^i \mid U^i \in (b_{k-1}, b_k], X^i = x\right].$$

Corollary 1 holds because the condition puts the same individuals in each cell of the reported rank and of the latent rank, so the two conditional expectations average over the same people. The condition is strictly weaker than $\hat{U}^i = U^i$. It permits arbitrary misranking within cells. It is also compatible with rounded reports (Corollary A2) provided heaps of reports do not straddle cell boundaries.

Mean-Zero Error in Treatment Probabilities. Assumption 6 restricts only the ranks of reported treatment probabilities. The *ATT*, the *ATU*, and the *P RTE* use the levels as weights, so they need a different restriction, random error in $\hat{G}$ that is mean zero and uncorrelated with the subjective individual treatment effect and with the error in outcome reports. With this restriction, Assumption 7, Propositions 3 and 4 hold with Assumptions 5 and 7 in place of zero measurement error.

Assumption 7 (Mean-Zero Reporting Error in Treatment Probabilities). *Reported treatment probabilities satisfy $\hat{G}^i = G^i + \zeta^i$, where ζ^i has mean zero and is uncorrelated with the *seaITE*^{*i*} and with the reporting error in each state-contingent outcome expectation: $\mathbb{E}[\zeta^i] = 0$, $\mathbb{E}[\zeta^i \text{seaITE}^i] = 0$, and $\mathbb{E}[\zeta^i \bar{\eta}_D^i] = 0$ for $D \in \{0, 1\}$.*

Each condition removes one term from a $\hat{G}$ -weighted average. Mean-zero error leaves $\mathbb{E}[\hat{G}^i]$ equal to $\mathbb{E}[G^i]$. Error uncorrelated with the *seaITE* leaves the weighted gain unchanged. Error uncorrelated with $\bar{\eta}_D^i$ keeps the two reporting errors from compounding.

The rank and level conditions on reported treatment probabilities, Assumptions 6 and 7, can fail independently. Systematic distortion that preserves order biases the *ATT* weights of Proposition 3 but leaves the *MTE* unaffected. Mean-zero random error leaves the weighted averages unchanged but not the estimated *MTE*.

Random Error in Treatment Probability Ranks. If reported treatment probabilities have random error in ranks, the measured *MTE* curve is a flattened version of the true curve. It moves toward the *ATE* and never outside the range of the true curve. Under Assumptions 2 and 6, any two individuals with the same X and V report the same treatment probability. Proposition A2 in Appendix A drops both assumptions. Individuals with the same X and V may then report different treatment probabilities because they hold different subjective beliefs or answer with error. Its hypotheses are Assumptions 1 and 3, a continuous distribution of $\hat{G}$ given X , and Assumption 5 with the measured rank in place of the treatment belief. The *ATE* implied by the measured curve is unchanged, and the heterogeneity of the measured curve is a lower bound on the true heterogeneity. Thus a sloped estimate is robust evidence of selection on gains, while a flat estimate is consistent with either true homogeneity or uninformative ranks.

Table 1: Which Results Require Which Assumptions?

Parameter	Survey objects	Belief conditions: $seaTE = TE$	Reporting conditions: $seaTE = seaTE$
ATE	$\hat{h}_D^i$	Assumption 3	Assumption 5
ATT , ATU , and other G -weighted averages	$\hat{h}_D^i$; levels of $\hat{G}^i$	Assumptions 3 and 4	Assumptions 5 and 7
$PRTE$	$\hat{h}_D^i$; levels of $\hat{G}^i$; each elicited in both regimes	Assumptions 3 and 4 in each regime; average bias in outcome-belief levels common across regimes	Assumptions 5 and 7 in each regime; C^i common across regimes
Binned MTE	$\hat{h}_D^i$; bins of $\hat{G}$ -ranks	Assumptions 1 and 2 for the rank; Assumption 3	Assumption 5; reported ranks correct up to the coarseness of the partition in place of Assumption 6 (Corollary 1)
MTE function	$\hat{h}_D^i$; ranks of $\hat{G}^i$	Assumptions 1 and 2 for the rank; Assumption 3	Assumptions 5 and 6

Notes: Each row lists the conditions under which the measured subjective ex ante parameter equals its objective counterpart, denoted TE . The rows are not nested: the ATT , ATU , and $PRTE$ rows use the levels of reported treatment probabilities and no rank condition, and the MTE rows use ranks and no condition on levels. Ex ante versions of the parameters throughout.

2.6 Relation to the Instrumental Variables Method

The standard approach to identifying MTE s uses instrumental variables. That method and ours are each organized around a latent selection rank, but they obtain identifying information in fundamentally different ways. The IV method learns about the treatment gain at a latent margin by exogenously moving that margin. Our method locates each individual on the latent rank using elicited treatment beliefs and pairs that rank with elicited treatment-contingent outcome expectations, identifying the $seaMTE$ directly. Under Assumption 3, the $seaMTE$ equals the MTE . No instrument is observed or assumed in our identification results.

The Standard IV Method. In the IV method, selection into treatment is modeled as

$$D^i = \begin{cases} 1 & \text{if } \mu(X^i, Z^i) \geq V^i \\ 0 & \text{if } \mu(X^i, Z^i) < V^i, \end{cases} \quad (15)$$

where $Z \in \mathbb{R}^{d_z}$ are observed instruments and $\mu(\cdot)$ is a deterministic function. The instruments must be (i) independent of the latent selection component conditional on covariates, $Z \perp V|X$, and (ii) excluded from potential outcomes, $\mathbb{E}[Y_D|X, Z, V] = \mathbb{E}[Y_D|X, V]$ for $D \in \{0, 1\}$. Vytlacil (2002) shows that this latent-index selection structure with condition (i) is equivalent to the independence and monotonicity assumptions of Imbens and Angrist (1994). Condition (ii) is the separate exclusion restriction that gives variation in the propensity score a causal interpretation for outcomes. The instruments generate the propensity score $P(x, z) := P(D^i = 1|X^i = x, Z^i = z) = F_{V|x}(\mu(x, z))$ and, with $U^i = F_{V|X^i}(V^i)$ as in Section 2.1, selection can be rewritten as $D^i = 1$ if and only if $P(X^i, Z^i) \geq U^i$. Heckman and Vytlacil (1999, 2001, 2005) show that the *MTE* is identified by the local instrumental variables (*LIV*) estimand: with Y^i the realized outcome defined after equation (9),

$$LIV(u, x) := \left. \frac{\partial \mathbb{E}[Y^i | X^i = x, P(X^i, Z^i) = p]}{\partial p} \right|_{p=u} = MTE(u, x) \quad (16)$$

at every u in the interior of the support of $P(X, Z)$ conditional on $X = x$. Exogenous variation in Z moves the treatment margin $P(X, Z)$ while conditions (i) and (ii) leave the distribution of U and the mean outcome functions $\mathbb{E}[Y_1|X, U]$ and $\mathbb{E}[Y_0|X, U]$ unchanged, so comparing average realized outcomes across nearby margins isolates the expected treatment gain of exactly those individuals whose rank lies at the margin. This comparison is the *LIV* derivative in equation (16). When its conditions are credible, this design identifies treatment effects from realized choices and outcomes, with no reliance on measuring what individuals believe.

The extent of identification depends on the support of the induced propensity score. Identifying the *MTE* function at every rank requires an instrument rich and powerful enough that, conditional on each value of X , the propensity score varies continuously over $(0, 1)$ (Carneiro, Heckman, and Vytlacil, 2011). Such instruments are rare. An instrument whose propensity score takes finitely many values identifies averages of the *MTE* over the intervals between those values. A continuous instrument with limited support identifies the *MTE* pointwise on the interior of that support and not outside it. The pointwise *MTE* function and treatment parameters that weight unsupported regions are at best partially identified (Mogstad, Santos, and Torgovitsky, 2018).

The Subjective Expectations Method. Our method replaces this indirect route with direct measurement, and the subjective treatment probability plays a different role than the instrument. G^i is the individual's own forecast of D^i , based on information that includes V^i , whereas $P(X, Z)$ is the best forecast available to the econometrician, who does not observe V^i .

Although $\hat{G}^i$ predicts treatment, it is not an instrument for treatment. It predicts treatment because it is informative about the latent type V^i , not because it shifts the treatment threshold exogenously. Because V^i may be related to Y_D^i , $\hat{G}^i$ may be associated with potential outcomes even though eliciting the report has no causal effect on outcomes. Thus relevance and the absence of a direct effect on outcomes do not make $\hat{G}^i$ a valid instrument. An instrument moves the treatment margin for otherwise comparable latent types, whereas $\hat{G}^i$ predicts treatment by sorting individuals across latent types. We use $\hat{G}^i$ as a measurement of the latent type. Under Assumptions 1, 2, and 6, the within- X rank of reported treatment probabilities equals U^i , so the econometrician conditions on the latent rank directly rather than generating exogenous variation around it. Three differences from the IV method follow.

First, no exclusion restriction is required. In our model, outcomes may depend directly on the selection shock μ . In the running example, the promotion affects both the woman's propensity to have a child and her propensity to work after childbirth. A shifter of this kind cannot serve as an instrument. It violates exclusion, and an *LIV* estimand built on it would confound the treatment effect with the direct effect of the shifter on outcomes. The survey method remains valid because each respondent's expectations integrate over her own uncertainty about μ , and the ex ante *MTE* is defined accordingly. Hence the survey method is most useful when salient determinants of selection also affect potential outcomes, so that no credible excluded shifter is available as an instrument.

Second, there is no support requirement when using $\hat{G}^i$. The within- X ranks of reported beliefs index the full support of U , so the *MTE* is identified over the entire unit interval. The binding requirement is instead on beliefs. Treatment must not be perfectly forecastable, so that reported probabilities are dispersed, interior, and informative about the latent rank (Assumption 1).

Third, in place of instrument validity we require the belief and reporting conditions of Assumptions 2, 5, and 6: a common forecasting rule, conditionally unbiased reports of expected outcomes, and rank-preserving reporting. The covariates X define the groups within which the common forecasting rule holds; no component of X need shift treatment or satisfy an exclusion restriction. The two sets of conditions share a structure. Assumption 2 imposes on beliefs the same single-index structure that, by the equivalence result just cited, monotonicity imposes on selection. Conditional on X , the subjective probability of treatment must move monotonically with the single latent factor V .

The two methods are alternative identification routes and complementary sources of evidence. Neither set of assumptions nests the other. The IV method uses realized choices and outcomes, so it does not rely on the correctness or reporting of beliefs. But it requires credible exogenous variation and realized outcomes, and it identifies the *MTE* only over

the margins the variation supports. The survey method identifies U^i and $seaMTEs$ before treatment or outcomes are realized and extends to counterfactual policy environments (Section 3.5). Its objective interpretation rests on Assumption 3, which cannot be verified from survey data alone. The methods can therefore discipline each other. Where a credible quasi-experimental design exists, it provides an external benchmark for treatment-effect functionals implied by the survey estimates. This is our strategy in Section 3.4.2, where we compare $ATTs$ computed from $\widehat{seaMTEs}$ with event-study estimates. Where instruments are discrete, weak, or unavailable, the survey method remains implementable under its own belief and reporting conditions and identifies the $seaMTE$ function at every rank.

3 The Effect of Childbirth on Female Labor Supply

We apply our method to study the effect of childbirth on labor supply, analyzing data from a Danish custom survey in which we elicit subjective beliefs concerning future childbirth and work. This demonstrates how to quantify the variables needed to estimate subjective ex ante $MTEs$ as described in Section 2.3. Furthermore, the survey responses can be matched to administrative records, which allows us to validate survey responses against historical behavior. This is particularly valuable because it allows us to compare treatment effects estimated using our survey methodology to those estimated using a traditional event-study method.

We present the institutional context and the survey, show how to compute $MTEs$ following Section 2.3, compare the survey responses with historical fertility and labor supply from administrative registries, and end with the counterfactual policy scenarios.

3.1 Institutional Context: Childcare and Parental Leave in Denmark

The institutional description in this subsection is as of May 2019, when the survey was fielded. The labor force participation rate of working-age women is about 80 percent. This is facilitated by a well developed and heavily subsidized high-quality public childcare system and by maternity leave policies that protect employment. Public day care is organized by municipalities and is the dominant type of day care provided. All children are eligible for day care unless a parent is on publicly funded maternity or parental leave. Opening hours are typically 6:30am to 5:00pm on weekdays, covering the entire work day throughout the work week for most people. Public day care is regulated by law, with the objective of ensuring childcare availability for all children and securing child development and learning. To ensure high quality, rules regulate the type and quantity of staffing, physical surroundings, safety, and

hygienic standards, and municipalities are required by law to monitor childcare institutions to ensure that the personnel have sufficient qualifications and that safety and hygienic standards are met. An important dimension of quality is staffing. In nurseries (ages 0–2) there are typically three children per employee, in kindergartens (ages 3–5) there are typically six, and about half of the employees have formal pedagogical training. Public day care is heavily subsidized, and users typically pay about one third of the total cost. The typical monthly user cost is about 3,000 DKK for a nursery slot and 1,700 DKK for a kindergarten slot (1 USD $\approx$ 7 DKK). The design of the public day care system has been in place since the 1970s, so childcare opportunities are well known and predictable. About 83 percent of all children aged 0–5 are enrolled in public or publicly funded day care.

Parents are entitled to up to 52 weeks of leave in connection with having a child. Eighteen weeks, starting four weeks before childbirth, are earmarked to the mother and two weeks are earmarked to the father. The remaining 32 weeks can be shared between parents as they prefer. The parental leave scheme guarantees job security and a minimum income corresponding to the unemployment insurance level. For a full-time employee this amounts to 90 percent of earnings and a maximum of about 19,000 DKK per month before taxes in 2020. These rules have been in place since 2002. We refer to Andersen (2018) for a more detailed description of the history of the parental leave system.

3.2 Survey and Registry Data

3.2.1 The Custom Survey

The survey elicits the subjective probability of having a child within the next four years, which is the probability of treatment, and the subjective probability of working at horizons $s = -12, -3, 3, 9, 18$, and 36 months relative to the birth, conditional on having a child. Because the outcome is binary, each reported probability of working is a state-contingent outcome distribution. These responses are the data objects of Section 2: the reported treatment probability, $\hat{G}^i$, and the reported child-contingent expected outcomes, $\hat{\mathbb{E}}^i [Y_{1,s}^i]$. Respondents answer every question on a slider with a random initial position. Appendix D reproduces the response screen and the fertility and labor supply questions.

To elicit fertility, we ask how likely the respondent is to try to get pregnant within the next four years, and then how likely she is to have a child if she does try and if she does not. The reported treatment probability compounds the three responses:

$$\hat{G}^i = \hat{p}^i(\text{try}) \hat{p}^i(\text{child} | \text{try}) + (1 - \hat{p}^i(\text{try})) \hat{p}^i(\text{child} | \text{not try}). \quad (17)$$

To elicit labor supply with a child, we ask the respondent to suppose that she does have a child within the next four years and to report the percent chance that she is working at each of the six horizons.¹¹ The question is asked separately for the cases in which her partner does and does not take at least two months of parental leave in the child’s first year. The child-contingent work expectation at each horizon compounds the two responses with the reported probability of partner leave:

$$\widehat{\mathbb{E}}^i [Y_{1,s}^i] = \hat{p}^i(\text{leave}) \hat{p}^i(\text{work}_s | \text{child, leave}) + (1 - \hat{p}^i(\text{leave})) \hat{p}^i(\text{work}_s | \text{child, no leave}). \quad (18)$$

The reported probability of working 12 months before the birth serves as the no-child expectation at every horizon, so the measured treatment effect at horizon s is

$$\widehat{seaITE}_s^i := \widehat{\mathbb{E}}^i [Y_{1,s}^i] - \widehat{\mathbb{E}}^i [Y_{1,-12}^i]. \quad (19)$$

This equals the $\widehat{seaITE}$ of equation (10) if, without a child, the respondent expects her probability of working to remain at the level she reports for 12 months before a birth. The $\widehat{seaMTE}$ needs less. The gap between the two expectations enters as the state-specific reporting error of Assumption 5, so the $\widehat{seaMTE}$ is the $seaMTE$ if the gap averages to zero within cells of observables and treatment probability. The same condition on realized labor supply, that absent a birth it would remain at its month -12 level, is the identifying assumption of the event study in Section 3.4.2.

3.2.2 Sampling and Link to Registry Data

The sample invited to participate in the Danish survey is recruited from the population registry. The population registry is a complete registry of all persons who are born or have ever had a registered address in Denmark. It contains the CPR-number, which is a personal identifier applied universally to record any contact an individual has with the public sector. For the survey, we were given access to a random sample of women aged 18–40 on December 31, 2018. The survey was fielded in May 2019. Invitations to participate were sent out using an official email account called *e-boks* that all Danes have. We sent out 51,000 emails and collected 11,304 responses.

In addition to the survey data we use third-party reported administrative register data compiled by Statistics Denmark from various government agencies. In particular, we use

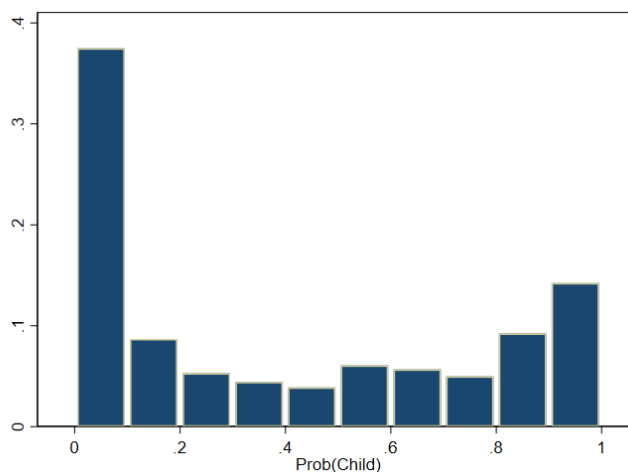
¹¹We interpret the “suppose” wording interventionally: the respondent imagines the birth state being set rather than learning from it, so each response is the state-contingent expectation of Section 2.3. This reading holds her uncertainty about μ^i and ϵ^i fixed across the two states. Appendix C analyzes the case in which the supposition instead informs her beliefs about μ^i .

the registries that document household composition, including how many children women have and when they were born. We also rely heavily on data from an administrative register, called the *e-income* registry, that contains employer-reported monthly wages and salaries for employees. All firms in Denmark have to report earnings and hours supplied for each employee, and hence whether the person is working, to the tax agency (SKAT). Since tax evasion on wage income is known to be very limited (Kleven, Knudsen, Kreiner, Pedersen, and Saez (2011)) we believe this data resource is accurate and precise. This registry has been in place since January 2008 and has previously been used for research purposes by Kreiner, Leth-Petersen, and Skov (2016) among others. We complement the registry with information about the receipt of public transfer income and with information about household composition and various background characteristics. Registry data are available for the entire population, and we merge the administrative data to the women we invited. Table A1 in Appendix E displays summary statistics for participants and non-participants. The table shows that participants are slightly older, slightly more likely to have a partner, have slightly more children, are less likely to be non-citizens, have slightly higher income and have slightly longer educations. While these differences are often statistically significant, in most cases the quantitative differences are modest.

3.3 Results: Estimating MTEs Using Subjective Expectations Data

3.3.1 Subjective Expectations Data

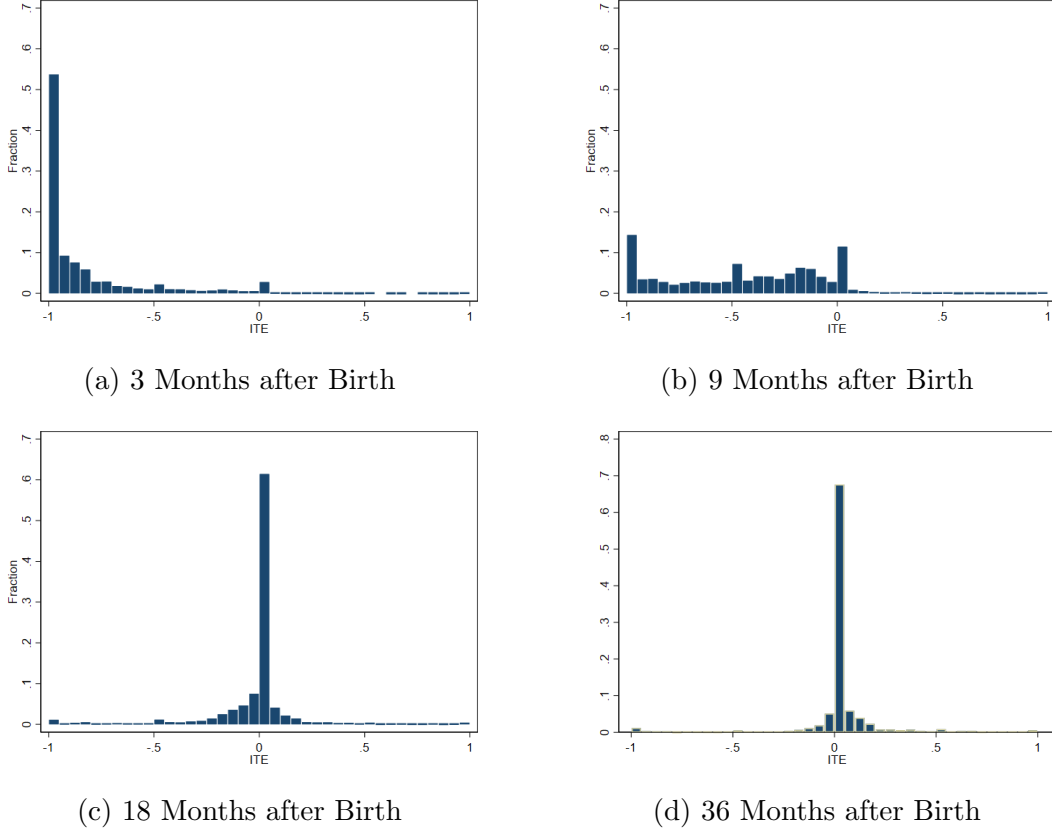
Figure 1: $\hat{G}^i$, The Reported Treatment Probability (Having a Child)



Notes: This figure plots a histogram of survey responses of the subjective probability of having a child over a 4-year horizon.

The probability of having a child is a key input for estimating *MTEs*. In Figure 1 we present

Figure 2: $\widehat{seaITE}_s^i$ Density



Notes: Each panel plots the histogram across respondents of $\widehat{seaITE}_s^i = \widehat{\mathbb{E}}^i[Y_{1,s}^i] - \widehat{\mathbb{E}}^i[Y_{1,-12}^i]$, the reported probability of working s months after a birth minus the reported probability of working 12 months before it, both conditional on having a child. The pre-birth report serves as the no-child expectation at every horizon (Section 3.2.1).

a histogram of the distribution of expected fertility within the four years following the survey. Childbirth expectations are distributed across the entire unit interval.¹²

The other key input is the subjective ex ante individual treatment effect, $\widehat{seaITE}_s^i$, the difference between a respondent's expected probabilities of working with and without a child. Figure 2 plots its density at four horizons. Three months after birth (Figure 2a) the $\widehat{seaITE}$ is concentrated around -1 , because most women expect to work without a child and few expect to work three months after a birth. Nine months after birth (Figure 2b) there is substantial heterogeneity. At 18 and 36 months (Figures 2c and 2d) almost all women expect the same probability of working with a child as without one. Because the MTE averages the $\widehat{seaITE}$ within rank cells, heterogeneity in the $\widehat{seaITE}$ bounds heterogeneity in the

¹²The histogram appears to indicate that roughly half of the women in the sample report extreme probabilities, i.e., 0 or 100. This is, however, due to the coarseness of the bins. In fact, 17 percent of the women in the sample indicate zero percent probability of childbirth and 9 percent of the women indicate 100 percent probability of childbirth.

MTE. Thus, there is scope for substantial *MTE* heterogeneity at 9 months, limited scope at 3 months, and very little at longer horizons. Whether the 9-month *MTE* actually varies depends on how the $\widehat{seaITE}$ covaries with the latent propensity to have a child.

Table A2 in Appendix G regresses the $\widehat{seaITE}$ at each horizon on characteristics measured in the registries in 2018, the year before the survey. The expected effect of childbirth on labor supply varies systematically with observables at the 3- and 9-month horizons. For example, women who already have children expect childbirth to reduce their probability of working by more at these horizons than women without children do, and more educated women expect larger reductions at all horizons. Reported outcome beliefs that were mere noise, uninformative about true beliefs, would not covary systematically with characteristics measured in the registries. We now examine whether the $\widehat{seaITE}$ varies systematically with the reported latent treatment propensity, $\hat{U}^i$.

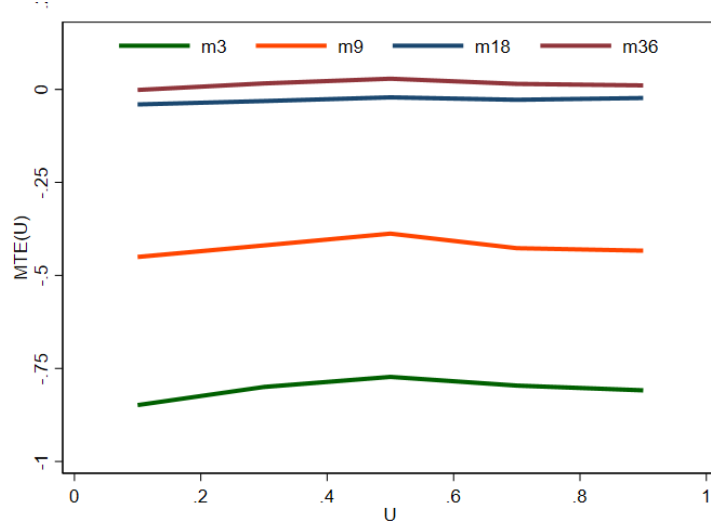
3.3.2 Estimating MTEs

Equation (11) suggests an estimator of *MTEs* that relies on the subjective probability data presented in Figures 1 and 2. In practice, we first select some X variables (e.g., age and number of children). Then we create a partition over X . For example, (age ≤ 30 , no child); (age > 30 , no child); (age ≤ 30 , 1 child); et cetera. Within each subset of the X partition, we order individuals according to $\hat{U}^i$ and then partition over $\hat{U}^i$, e.g., quintiles (Corollary 1). Then, for each subset of the $\hat{U} \times X$ partition, we compute $\widehat{seaMTE}(u, x)$ as the average $\widehat{seaITE}$ for members of that subset. Once we compute the $\widehat{seaMTE}$ s, we can compute $\widehat{ATE}$ s, $\widehat{LATE}$ s, $\widehat{ATT}$ s, and $\widehat{ATU}$ s by appropriately weighting by $\hat{G}^i$, following the weighting logic of Heckman and Vytlacil (2005). Proposition 3 states the conditions under which the resulting averages recover the ex ante parameters. For example, the *ATE* is just the unconditional average $\widehat{seaITE}$. The *ATT* is the weighted average of the $\widehat{seaITE}$ s, with more weight on people more likely to be treated: $\left(\omega^i = \frac{\hat{G}^i}{\text{mean}(\hat{G}^i)}\right)$.

Figure 3 plots the estimated $\widehat{seaMTE}_s$ at each horizon, ranking all women together without partitioning X .¹³ Respondents expect an 80–90 percentage point reduction in the probability of working 3 months after a birth, a 50 percentage point reduction after 9 months, and very modest effects after 18 and 36 months. The curves are mostly flat, with a slight slope at months 3 and 9. Limited selection on unobserved factors is plausible in a small and homogeneous society such as Denmark, where a high level of income redistribution, large subsidies for publicly provided high-quality childcare, and universal job protection laws reduce heterogeneity in opportunity costs.

¹³In Figure 3 we have binned $\hat{U}$ into five equally spaced groups. The results are robust to binning into ten groups. To avoid clutter we have omitted standard errors. In all cases, standard errors are very small.

Figure 3: Subjective Ex Ante Marginal Treatment Effects ($\widehat{seaMTE}_s$)

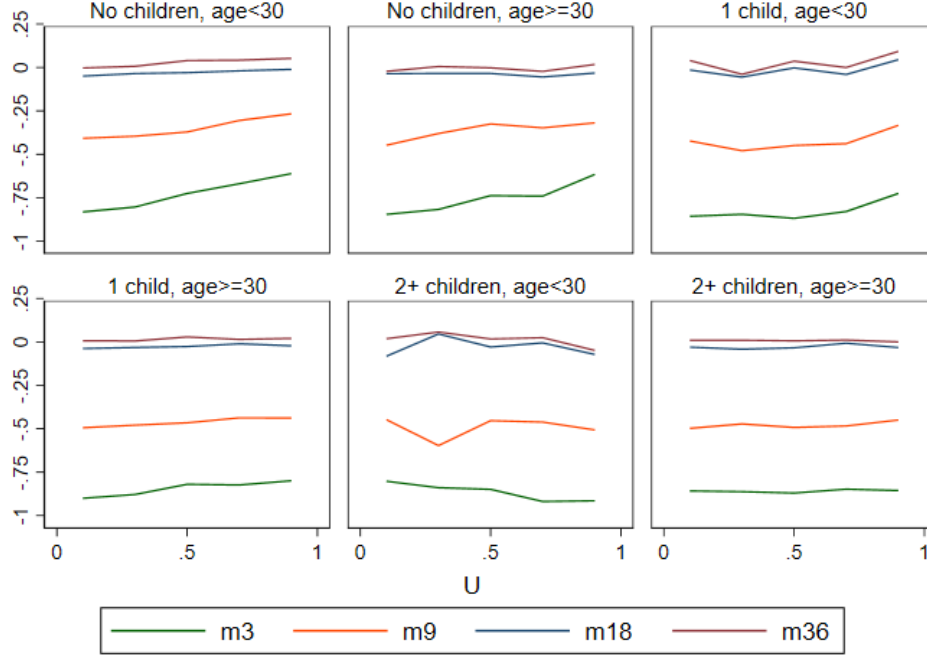


Notes: Figure 3 shows marginal treatment effects calculated for various horizons after childbirth. We order individuals by their $\hat{U}^i$ and calculate the average $\widehat{seaITE}$ within five equally sized buckets of $\hat{U}^i$.

Ranking without conditioning on X , as in Figure 3, imposes Assumption 2 unconditionally, so differences in reported probabilities across women with different observables are attributed to the latent component V rather than to X . The $seaMTE$ may also vary with X , and ranking on $\hat{U}$ alone would mask this heterogeneity. Thus, we explore the $\widehat{seaMTE}(u, x)$ function by ranking within cells of a covariate partition for a few X variables that the literature associates with working after childbirth. The chosen X s are economically interesting and also demonstrate how to estimate $seaMTE(u, x)$ nonparametrically. Fully nonparametric double partitioning (X cells crossed with rank bins) is demanding of the data, which limits how rich a covariate set can be used. With richer conditioning, semiparametric implementations are natural refinements, for example residualizing $\widehat{seaITE}$ and $\hat{G}$ on X before ranking or specifying $\widehat{seaMTE}(u, x)$ as additive in u and x . Because the purpose of the application is to demonstrate the method transparently, we retain the nonparametric implementation.

Figure 4 shows that the main source of selection on unobservables is at the 3-month horizon among women with no other children. The positive slope of $\widehat{seaMTE}(u, x)$ at 3 months for these women means that those less likely to have a child expect a smaller drop in their probability of working 3 months after a birth. Otherwise, the $\widehat{seaMTE}$ s are remarkably constant across older and younger women, across women with 1, 2, or 3 other children, and across all women at longer horizons.

Figure 4: Subjective Marginal Treatment Effects Conditional on Age and Number of Children



Notes: Figure 4 shows marginal treatment effects calculated for various horizons after childbirth. We rank $\hat{U}^i$ within elements of the partition of the Age and Number of Children X variables and subsequently calculate the average $\widehat{seaITE}$ within five equally sized buckets of $\hat{U}^i$ for each subset of the partition.

3.4 Validating the Survey Method: Comparison to Historical Data

We now compare fertility and labor supply beliefs with historical data from administrative registries. We show that the average treatment effect on the treated (ATT) estimated using the survey data equals that estimated from a traditional event-study approach. We view this as supportive evidence for the level and outcome-belief conditions that underlie our average treatment effect estimates (Proposition 3) and, more tentatively, as evidence that $\widehat{seaMTE} = seaMTE = MTE$ in this context.

3.4.1 Subjective Beliefs and Realizations of Childbirth have Similar Conditional Correlations with Key Demographic and Economic Variables

The average stated expected probability of having a child in the survey is 40 percent. To compare this number with historical fertility, we consider all women aged 20–40 in 2012 and follow them four to six years forward. For this group, average fertility is 29 percent after four years, 35 percent after five years, and 41 percent after six years. We take this to reflect that stated expected fertility aligns reasonably well with fertility realized historically.

Table 2: **Regression of Historical and Expected Fertility on Covariates**

	(1) historical prob(child)	(2) subjective prob(child)
1 child	-0.117*** (0.014)	-0.169*** (0.011)
2 children	-0.459*** (0.002)	-0.278*** (0.012)
3 children	-0.472*** (0.002)	-0.320*** (0.025)
Educ, high school	-0.033*** (0.002)	-0.088*** (0.018)
Educ, vocational	0.075*** (0.002)	-0.022 (0.015)
Educ, middle	0.152*** (0.003)	0.027* (0.015)
Educ, college	0.169*** (0.003)	0.067*** (0.015)
Age -25	-0.018*** (0.003)	0.015 (0.015)
Age 31-35	-0.050*** (0.002)	-0.027** (0.012)
Age 36-40	-0.325*** (0.002)	-0.250*** (0.012)
Partner	0.334*** (0.002)	0.119*** (0.010)
Income -200	-0.047*** (0.002)	0.005 (0.012)
Income, 200-300	0.006** (0.002)	0.021* (0.012)
Income, 400-500	-0.020*** (0.003)	-0.010 (0.013)
Income, 500-	-0.045*** (0.004)	-0.046*** (0.016)
Constant	0.439*** (0.003)	0.450*** (0.018)
N	725,263	6,898

Notes: Column (1) presents a regression of the probability of having a child during 2013–2018 for women aged 20–40 in 2012 according to the population registry. Regressors are measured in 2012. Column (2) presents a corresponding regression where the dependent variable is expected fertility in the baseline scenario from the survey. The regressors are from administrative registries and are measured in 2018. Robust standard errors in parentheses.* 10%, ** 5% *** 1% level of significance.

In Table 2 we regress fertility on demographic and economic characteristics. In column (1) the dependent variable is realized fertility during 2013–2018 for all women aged 20–40 in 2012 and characteristics are measured in 2012. In column (2) the dependent variable is stated expected fertility and characteristics are measured in the registry data in 2018. Considering first the results for the survey sample in column (2), expected fertility is negatively correlated with the number of children. Fertility is positively correlated with the level of education and is lower for women older than 30. Having a partner is positively correlated with expected fertility. Given the other covariates, income is uncorrelated with expected fertility except in the highest bracket, where the correlation is negative.

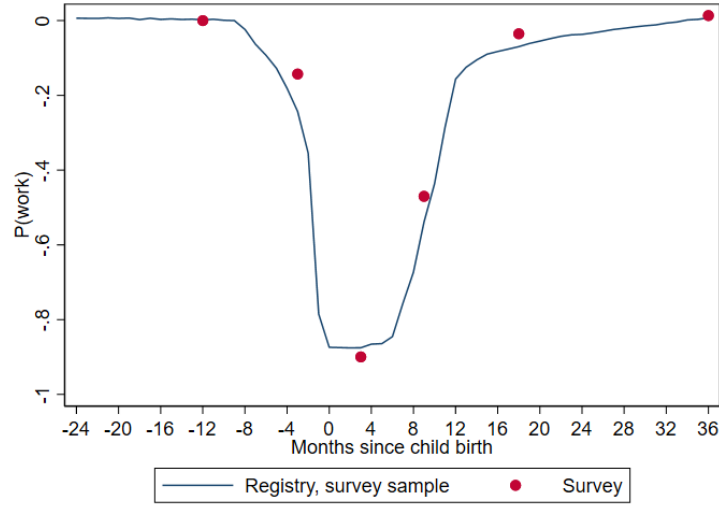
In column (1) we find much the same pattern, although in some cases the estimated parameters differ in size and standard errors are smaller due to the large sample size. It is not possible with these data to verify directly that $\hat{G}^i$ preserves the rank of G^i , as Assumption 6 requires. However, that the estimates in columns (1) and (2) of Table 2 generally share signs, and often magnitudes, supports the view that fertility expectations rank women similarly to their actual fertility chances.

3.4.2 Subjective Expectations *ATTs* and Event-Study *ATTs* Align at Every Horizon

Figure 5 displays with red dots the average treatment effects on the treated (*ATTs*) estimated using subjective probability data as detailed in Section 3.3.2. The red dots mark the expected percentage of women working at different horizons around childbirth, where the fraction of women working in period -12 is set to zero, so that estimates for other periods give the deviation in labor supply from the level at month -12 . Expected labor supply starts decreasing three months before childbirth, and most women expect to be out of work three months after the birth. Expected labor supply does not drop to -1 because not all women participating in the survey expect to be working 12 months before the expected childbirth. Nine months after childbirth the average probability of working is about 50 percentage points below the level in month -12 . Eighteen months after birth average expected labor supply is only about 3 percentage points lower than at the outset. Three years after childbirth labor supply has returned to the same level as before pregnancy.

To assess whether labor supply expectations around childbirth are likely to reflect real behavior, we compare survey responses with historical labor supply behavior. The comparison sets the subjective expectations *ATTs* against event-study *ATTs* estimated on registry data.

Figure 5: Estimating Average Treatment Effect on the Treated (*ATTs*): Subjective Expectations Method compared to Historical Event Study



Notes: The figure shows the percentage of women working at different horizons relative to childbirth, where the fraction of women working in period -12 is set to zero. Estimates thus give the deviation in labor supply from the level at month -12. The red dots show the survey *ATT*, the average across respondents of the $\widehat{seATE}_s^i$ of Section 3.2.1, weighting by the relative probability of having a child. The solid blue line shows event study *ATT* estimates, i.e., δ from equation (20), based on childbirths during the period 2010–2015 recorded in the administrative registries for the women who are in the survey sample.

To do this we identify the survey respondents who gave birth during 2010–2015.¹⁴ For these childbirths we follow the labor supply of the mother at a monthly frequency from 24 months before to 36 months after childbirth. To quantify the labor supply response around childbirth we employ an event study design by estimating the following regression:

$$y_{it} = \sum_{k \in K} D_{it}^k \delta_k + \gamma X_{it} + u_{it}. \quad (20)$$

Here y_{it} is a dummy variable equal to one if individual i is working in month t , D_{it}^k is a dummy for the child being born k periods ago, and δ_k measures the effect of the childbirth k periods around birth. The month of conception is omitted, so $k \in K = \{-24, -23, \dots, 36\} \setminus \{-9\}$. X is a vector of dummy variables controlling for the age of the mother and the timing of adjacent children that the mother has also given birth to. Women

¹⁴See Appendix F.2 for analysis using all women recorded in the *e-income* registry. Results are very similar, although expected labor supply is slightly higher than historical labor supply at longer horizons in the full population. That the survey sample estimate is closer to expected labor supply could reflect a modest level of self-selection into participating in the survey, such that women who are more likely to return to work after giving birth are more likely to participate. Given the small deviations, we conclude that such self-selection is moderate.

who give birth to more than one child during the period 2010–2015 enter with a separate event time line for each child.¹⁵ The estimates of δ_k measure the change in labor supply k months around childbirth relative to the omitted month. Because equation (20) is estimated on a sample of women who all have children, but at different points in time, δ_k can be interpreted as an estimate of the average treatment effect on the treated (*ATT*). Thus, these event-study *ATT*s are directly comparable to the *ATT*s computed from estimated *MTE*s presented in Figure 5.

Two clarifications about the comparison. First, both series are treated labor-supply paths relative to the pre-birth month -12 . The survey series is the $\hat{G}^i$ -weighted average of the $\widehat{seaITE}_s^i$, which subtracts the month -12 report respondent by respondent, so it equals the child-contingent path weighted by $\hat{G}^i$ and normalized at $s = -12$. The event-study series is built from realized outcomes of mothers and renormalized so that the month -12 coefficient is zero (the regression itself omits the month of conception, $k = -9$). Second, interpreting the common path as an *ATT* requires that, absent childbirth, labor supply would have remained at its month -12 level after netting out the controls. This is the standard event-study identifying assumption. Section 3.2.1 imposes the same condition on beliefs to make the $\widehat{seaITE}$ a treatment effect, so the two series rest on one economic condition and differ in whether beliefs or realizations supply the treated path. The event-study estimate is therefore the natural benchmark for the survey series.

In Figure 5 we plot estimates of δ_k . The event-study *ATT* nearly coincides with the survey *ATT* at every horizon in the survey population. We interpret this close alignment as substantial supportive evidence that $\widehat{seaMTE} = seaMTE = MTE$ in this context.

Which assumptions does this comparison test? By Proposition 3 and Table 1, the survey-based *ATT* uses the levels of reported treatment probabilities as weights, together with the elicited outcome expectations, and not the ranks. The close alignment of the two *ATT* series is therefore direct evidence on the level conditions, that the levels of reported treatment probabilities are right on average (Assumptions 4 and 7) and that the reported expected gain from treatment is correct (Assumptions 3 and 5). These conditions underlie every $\hat{G}$ -weighted average we report. The alignment is not, by itself, direct evidence on the rank conditions, Assumptions 2 and 6, which only the *MTE* function requires (Section 2.4). Still, we find it unlikely that the two *ATT*s would align so closely by coincidence if elicited beliefs were substantially distorted, because the survey *ATT* aggregates the same individual-level beliefs from which the $\widehat{seaMTE}$ function is built. Moreover, under the maintained assumptions,

¹⁵Appendix F contains further details about estimation. Estimation of equation (20) by OLS could be biased if treatment effects vary across treatment cohorts. In Appendix F.1 we present estimates based on the Sun and Abraham (2021) estimator that allows for such treatment effect heterogeneity. The results are practically identical to those presented in Figure 5.

the survey ATT is a weighted average of the $\widehat{seaMTE}$ function and the event-study ATT is a weighted average of the objective MTE function. Observing the same weighted average across the two approaches is therefore supportive, though not conclusive, evidence that the $seaMTE$ and MTE functions themselves coincide in this context.

3.5 Counterfactual Analysis using $seaMTEs$

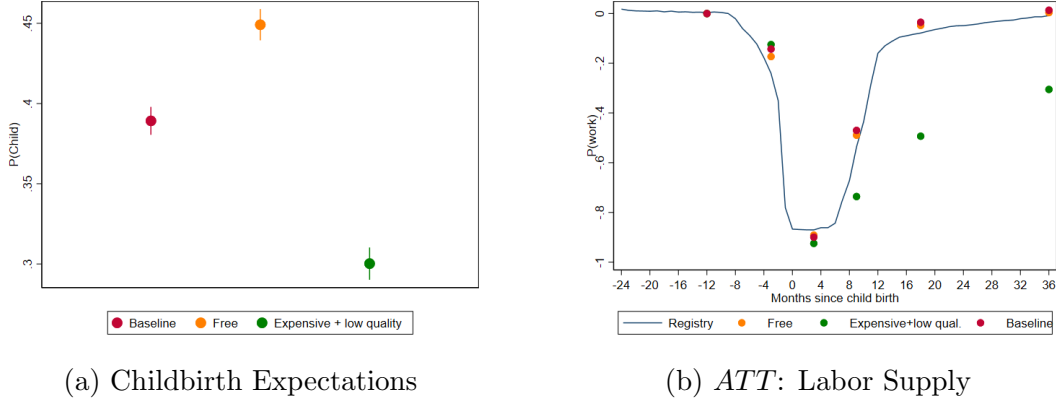
Of course, no method can validate counterfactuals about policies that have never been observed without further assumptions. However, in the absence of a valid and relevant experiment, which is often the case for large-scale policy interventions, counterfactual analysis using $seaMTEs$ provides a data-driven methodology to forecast the effects of a counterfactual policy. This method can complement predictions of pure theory by bringing data to the analysis and it can complement structural models by relying less on extrapolation of model predictions beyond the situations that informed the models' quantification.

To illustrate this, we ask about two counterfactual policies, one that makes childcare free and one that triples the price of childcare while also lowering its quality. The first policy is a relatively small perturbation of the existing Danish childcare system, while the second is a more drastic change that moves childcare in Denmark closer to childcare in the US. The policy scenarios are presented to respondents as follows:

- The first change we would like to ask you to consider is that as of today and in the foreseeable future, all municipal childcare is free. Assume that childcare quality is not affected by this cost change.
- The next change that we would like you to consider is that starting today and in perpetuity public child care is **low quality**. In particular, assume that child care staff was cut by 50% and that all employees are non-skilled. This would mean that no trained teachers would be employed, and that child care would focus on providing a place where the child can stay during work hours but not on child development. In addition, suppose that starting today and in perpetuity public child care was **more expensive**. In particular, suppose that the price of child care increased to be three times what it costs today. This would mean that the typical price of nursery care would increase to about 10,000 DKK per month for each child but that the quality is low as described above.

Within these alternative scenarios we ask about expected fertility and labor supply in exactly the same way as in the baseline case.

Figure 6: Counterfactual Policy Scenarios



Notes: Panel a shows average expected fertility in the baseline scenario as well as in two counterfactual policy scenarios. In one scenario childcare is made free and in the other scenario the price of childcare is tripled and the quality is reduced compared to the baseline. Panel b shows the average expected probability of working conditional on having a child for the baseline scenario and for the two counterfactual policy scenarios, weighting each respondent by her relative probability of having a child in that scenario ($\hat{G}^i$ divided by its scenario mean), as in Figure 5. The solid line shows event study estimates based on childbirths recorded in the administrative registries for the women who are in the survey sample.

Figure 6 presents average expected fertility and labor supply for the baseline and for the two counterfactual scenarios.¹⁶ The free childcare scenario is shown with yellow markers and the scenario with low quality and high price with green markers. Making childcare free increases expected fertility by about 5 percentage points and leaves labor supply conditional on having a child practically the same as in the baseline scenario. In contrast, lowering childcare quality and at the same time tripling the price has a dramatic influence. Expected fertility drops by 10 percentage points compared to the baseline scenario, and expected labor supply conditional on having a child is persistently lower, even after the job security and parental leave period runs out 12 months after birth. Eighteen months after birth it is 51 percentage points lower than the pre-birth level, and after 36 months it is still 32 percentage points lower.

Table 3 regresses fertility expectations and $\widehat{seaITEs}$ from the expensive, low-quality childcare scenario on covariates.¹⁷ In column (1) stated expected fertility in the expensive and low quality scenario is regressed on characteristics measured in the registry data in 2018. Qualitatively, the results are comparable to those for expected fertility in the baseline case in column (2) of Table 2. In columns (2)–(5) the dependent variable is the $\widehat{seaITE}$ for months

¹⁶Appendix H reports the distribution across respondents of the change in reported fertility probability between the baseline and each scenario.

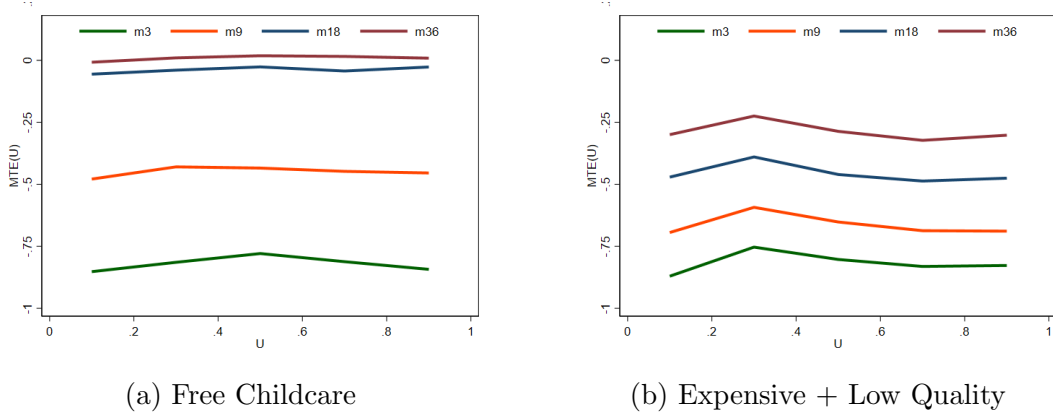
¹⁷Regressions for the baseline scenario corresponding to columns (2)–(5) of Table 3 are shown in Appendix G. There are fewer observations in the analysis of the counterfactual policy scenarios than in the baseline analysis because of item nonresponse. We have reproduced the baseline analysis using the subsample with valid responses to the questions about the counterfactual policy and the baseline results are practically identical to those presented in Figure 5.

Table 3: Counterfactual Scenario with Expensive and Low Quality Childcare

	(1) prob(child) b/se	(2) Month 3 b/se	(3) Month 9 b/se	(4) Month 18 b/se	(5) Month 36 b/se
1 child	-0.204*** (0.012)	-0.004 (0.011)	-0.045*** (0.014)	-0.070*** (0.015)	-0.044*** (0.014)
2 children	-0.262*** (0.014)	-0.008 (0.016)	-0.038* (0.020)	-0.078*** (0.022)	-0.074*** (0.022)
3 children	-0.235*** (0.034)	0.071 (0.045)	0.016 (0.050)	-0.075 (0.054)	-0.105** (0.053)
Educ, high school	-0.075*** (0.022)	-0.107*** (0.025)	-0.131*** (0.026)	-0.118*** (0.028)	-0.070*** (0.026)
Educ, vocational	-0.045*** (0.017)	-0.115*** (0.020)	-0.069*** (0.022)	-0.008 (0.023)	0.004 (0.021)
Educ, middle	0.002 (0.018)	-0.140*** (0.020)	-0.181*** (0.022)	-0.135*** (0.024)	-0.113*** (0.022)
Educ, college	0.064*** (0.018)	-0.137*** (0.020)	-0.156*** (0.022)	-0.113*** (0.023)	-0.057*** (0.021)
Age -25	-0.024 (0.019)	0.018 (0.019)	0.039* (0.021)	0.047** (0.022)	0.070*** (0.021)
Age 31-35	-0.050*** (0.014)	0.044*** (0.012)	0.040*** (0.015)	0.033** (0.017)	0.033** (0.016)
Age 36-40	-0.205*** (0.013)	0.033*** (0.012)	0.017 (0.015)	-0.005 (0.017)	0.006 (0.016)
Partner	0.075*** (0.012)	-0.015 (0.012)	-0.025* (0.014)	-0.087*** (0.016)	-0.046*** (0.014)
Income -200	-0.025* (0.014)	0.131*** (0.014)	0.103*** (0.017)	0.061*** (0.018)	0.027 (0.017)
Income, 200-300	-0.001 (0.015)	0.040*** (0.013)	0.006 (0.017)	-0.006 (0.019)	0.002 (0.018)
Income, 400-500	-0.008 (0.016)	0.003 (0.013)	-0.005 (0.017)	0.002 (0.019)	0.016 (0.019)
Income, 500-	-0.032* (0.019)	-0.020 (0.016)	0.005 (0.021)	0.032 (0.024)	0.038* (0.022)
Constant	0.422*** (0.021)	-0.763*** (0.022)	-0.564*** (0.025)	-0.317*** (0.027)	-0.217*** (0.025)
N	5,088	4,961	4,954	4,947	4,959

Notes: Column (1) presents a corresponding regression where the dependent variable is expected fertility in the counterfactual scenario where childcare is expensive and low quality. The regressors are from administrative registries and are measured in 2018. Columns (2)-(5) contain regressions where the dependent variable is the $\widehat{seATE}_s^i$ for $s = 3, 9, 18$, and 36 months after childbirth. Robust standard errors in parentheses. * 10%, ** 5% *** 1% level of significance.

Figure 7: $\widehat{seaMTE}_s$ in Counterfactual Policy Scenarios



Notes: Panel a shows marginal treatment effects calculated for various horizons after childbirth for the counterfactual policy scenario where childcare is made free. Panel b shows marginal treatment effects calculated for various horizons after childbirth for the counterfactual policy scenario where the price of childcare is tripled and quality is lowered. In each scenario we order individuals by their $\hat{U}^i$, computed from the treatment probability reported under that scenario, and calculate the average $\widehat{seaITE}$ within five equally sized buckets of $\hat{U}^i$, as in Figure 3.

3, 9, 18, and 36 after childbirth. Unlike in the baseline case, where heterogeneity across observables is concentrated at the 3- and 9-month horizons, there is considerable response heterogeneity at every horizon, including 18 and 36 months after birth. The largest expected declines in labor supply documented in Figure 6 are reported by women who already have children and by those with higher education.

Figure 7 plots the $\widehat{seaMTE}_s$ for the two counterfactual scenarios, Figure 7a for the free childcare scenario and Figure 7b for the scenario with high price and low quality. The $\widehat{seaMTE}_s$ under free childcare are practically identical to the baseline $\widehat{seaMTE}_s$ (Figures 7a and 3). Moreover, Figure 6a shows that making childcare free increases expected fertility by 5 percentage points. Under free childcare, then, the $MTEs$ are unchanged and the behavioral response is entirely an adjustment in the propensity to have a child. Making childcare free may be considered a relatively small policy change, because in the baseline childcare is already heavily subsidized, with 2/3 of the cost covered by the local government. Making childcare low quality and at the same time tripling the cost is a more dramatic change. In Figure 7b, the $MTEs$ under this policy are markedly lower than the baseline $MTEs$ shown in Figure 3.

Our counterfactual analysis based on $\widehat{seaMTE}_s$ connects to the policy relevant treatment effect ($PRTE$) of Heckman and Vytlacil (2005), defined in Section 2.4, which can be constructed from baseline $MTEs$ only if the baseline $MTEs$ are policy invariant. In our application, policy invariance requires that an alternative policy change only the propensity to have a child and not expected labor supply in either child state, and hence not the labor supply response to having a child. The policy invariance of estimated baseline $MTEs$ is typically difficult to test,

because naturally occurring data most often do not document behavior in alternative policy environments. An advantage of custom surveys is that they can ask about the probability of treatment as well as state-contingent outcomes in counterfactual policy scenarios. Proposition 4 formalizes this point. With beliefs elicited under each policy regime, the *seaP RTE* is identified without imposing policy invariance (part (i)), and it equals the *P RTE* under the belief conditions of part (ii). The fertility and labor supply changes in Figure 6 are changes in expectations. By Proposition 4(iii), they also forecast realized outcomes under each policy when, within type, having a child carries no information about either potential outcome. The alignment in Figure 5 supports that condition for the child state in the baseline regime. For the expensive, low-quality childcare counterfactual, the baseline *MTEs* are not policy invariant, so its effects cannot be computed from *P RTEs* based on baseline estimates.

4 Conclusion

This paper develops a nonparametric method that identifies *MTEs* using survey data on subjective expectations instead of instruments. The subjective treatment probability identifies the latent determinant of selection into treatment if people with the same observable characteristics use a common model to forecast their selection. Combined with subjective expectations about treatment-contingent outcomes, it identifies subjective marginal treatment effects. These equal objective *MTEs* when subjective expected outcomes are correct on average. We also show how purpose-designed survey data joined with historical administrative data test these assumptions, and how eliciting beliefs under counterfactual policies delivers the effects of policies before they are implemented.

We apply the method with childbirth as the treatment and female labor supply in Denmark as the outcome, using a custom survey linked to administrative registries. The estimated *MTEs* are roughly constant across levels of fertility beliefs, so after conditioning on typical demographic and economic observables there is little selection into having children based on a latent propensity to work after having a child. Having a child reduces the likelihood of working by about 50 percentage points 9 months after birth, around 3 percentage points 18 months after, and essentially zero 36 months after.

The average treatment effect on the treated computed from the estimated *seaMTEs* is nearly equal to the *ATT* estimated from realized labor supply with an event-study design. We view this alignment as evidence that the identifying assumptions hold and that subjective and objective *MTEs* are equal in this context.

Making childcare free leaves the labor supply path around a birth and the *MTE* function unchanged, consistent with policy invariance of the baseline *MTEs*. Tripling its price and

lowering its quality lowers expected fertility by about 10 percentage points and the expected probability of working 36 months after birth by 32 percentage points, and shifts the *MTE* function down, so baseline *MTEs* are not policy invariant.

Estimating *seaMTEs* complements existing methods where the treatment or outcomes have not yet materialized or where no credible instrument exists. Because beliefs can be elicited under a proposed policy, the method also measures the policy’s effect on outcomes while accounting for its effect on selection into treatment.

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Appendix

A Proofs

Propositions 1', 2', 3', and 4' restate Propositions 1, 2, 3, and 4 with Assumptions 5, 6, and 7 in place of zero measurement error. Each primed result implies its unprimed one, because zero measurement error implies the three assumptions (with $\hat{G}^i = G^i$, $C^i(y) = 0$, $\eta_D^i(y) = 0$, and $\zeta^i = 0$). Proposition A1 gives $\widehat{seaMTE} = seaMTE$ with Assumptions 5 and 6 in place of zero measurement error, under which the equality holds by definition; it is the measurement step of Proposition 2'. Corollary A1 states Proposition 2' under rational expectations, Corollary A2 states set identification of the latent rank under rounded reports, and Proposition A2 characterizes the measured MTE curve under random error in ranks.

Proposition 1' (Latent Treatment Propensity with Reporting Error). *Let Assumptions 1, 2, and 6 hold. Then $U^i = \tilde{U}^i = \hat{U}^i$.*

Recall the definitions $U^i := F_{V|X^i}(V^i)$, $F_{V|X^i}(v) := P(V \leq v | X = X^i)$, G^i the subjective probability of treatment, $\tilde{U}^i := 1 - F_{G|X^i}(G^i)$, and $\hat{U}^i := 1 - F_{\hat{G}|X^i}(\hat{G}^i)$.

Assumption 2 (Common Forecasting Rule). *For each x in the support of X there exists a CDF $\tilde{F}_{\mu|x}$, common to all individuals with $X^i = x$, such that $G^i = 1 - \tilde{F}_{\mu|x}(V^i)$.*

Assumption 6 (Rank-Preserving Treatment Probability Reports). *Suppose that $\hat{G}^i$ is reported such that the order of individuals equals their order in the true subjective probabilities. That is, $G^i < G^j \iff \hat{G}^i < \hat{G}^j \quad \forall i, j$.*

Proof. First, $\hat{U}^i = \tilde{U}^i$. By Assumption 6, reports preserve the order of true subjective probabilities, so there is a strictly increasing function ψ such that $\hat{G}^i = \psi(G^i)$ for all i .¹⁸ Since ψ is strictly increasing, the events $\{\psi(G) \leq \psi(G^i)\}$ and $\{G \leq G^i\}$ coincide, so

$$F_{\hat{G}|X^i}(\hat{G}^i) = P(\psi(G) \leq \psi(G^i) | X^i) = P(G \leq G^i | X^i) = F_{G|X^i}(G^i),$$

and therefore $\hat{U}^i = 1 - F_{\hat{G}|X^i}(\hat{G}^i) = 1 - F_{G|X^i}(G^i) = \tilde{U}^i$.

Second, $\tilde{U}^i = U^i$. By Assumption 2, $G = 1 - \tilde{F}_{\mu|X}(V)$ with $\tilde{F}_{\mu|X}$ a CDF and hence weakly increasing, so

$$F_{G|X^i}(G^i) = P(\tilde{F}_{\mu|X^i}(V) \geq \tilde{F}_{\mu|X^i}(V^i) | X^i) = P(V \geq V^i | X^i) = 1 - F_{V|X^i}(V^i),$$

¹⁸ ψ is well defined because Assumption 6 also implies that $G^i = G^j \Rightarrow \hat{G}^i = \hat{G}^j$: if instead $\hat{G}^i < \hat{G}^j$ (or $>$), then Assumption 6 would require $G^i < G^j$ (or $>$), a contradiction.

where the second equality holds because the two events differ only on the event that $\tilde{F}_{\mu|X^i}(V) = \tilde{F}_{\mu|X^i}(V^i)$ with $V < V^i$, which is contained in $\{G = G^i\}$ and therefore has probability zero because G is continuously distributed conditional on X (Assumption 1). The third equality uses $P(V = V^i|X^i) = 0$, which holds because V is continuously distributed conditional on X (Assumption 1). Therefore $\tilde{U}^i = 1 - F_{G|X^i}(G^i) = F_{V|X^i}(V^i) = U^i$. $\square$

Proposition A1 (*seaMTE with Reporting Error*). *Let Assumptions 5 and 6 hold. Then $\widehat{seaMTE}(u, x) = seaMTE(u, x)$.*

Recall the definitions of the subjective ex ante *ITE* and *MTE* and their measured counterparts:

$$\begin{aligned} seaITE^i &:= \int y [h_1^i(y) - h_0^i(y)] dy, & seaMTE(u, x) &:= \mathbb{E} [seaITE^i | \tilde{U}^i = u, X^i = x], \\ \widehat{seaITE}^i &:= \int y [\hat{h}_1^i(y) - \hat{h}_0^i(y)] dy, & \widehat{seaMTE}(u, x) &:= \mathbb{E} [\widehat{seaITE}^i | X^i = x, \hat{U}^i = u]. \end{aligned}$$

Assumption 5 (Conditionally Unbiased Outcome Reports). *For all i , for $D \in \{0, 1\}$, and for all y , reported beliefs satisfy*

$$\begin{aligned} \hat{h}_D^i(y) &= h_D^i(y) + C^i(y) + \eta_D^i(y), \\ \mathbb{E} [\eta_D^i(y) | X^i, G^i] &= 0. \end{aligned}$$

Proof. Throughout we assume all integrals are finite, so that the order of integration and expectation can be exchanged (Fubini's theorem). Substituting Assumption 5 into the definition of $\widehat{seaITE}^i$, the common error $C^i(y)$ cancels:

$$\begin{aligned} \widehat{seaITE}^i &= \int y [\hat{h}_1^i(y) - \hat{h}_0^i(y)] dy \\ &= \int y [(h_1^i(y) + C^i(y) + \eta_1^i(y)) - (h_0^i(y) + C^i(y) + \eta_0^i(y))] dy \\ &= seaITE^i + \int y [\eta_1^i(y) - \eta_0^i(y)] dy. \end{aligned}$$

Substituting this expression for $\widehat{seaITE}^i$ into the definition of $\widehat{seaMTE}(u, x)$ yields

$$\begin{aligned}
\widehat{seaMTE}(u, x) &= \mathbb{E} \left[\widehat{seaITE}^i \mid X^i = x, \hat{U}^i = u \right] \\
&\quad + \int y \mathbb{E} \left[\eta_1^i(y) - \eta_0^i(y) \mid X^i = x, \hat{U}^i = u \right] dy \\
&= \mathbb{E} \left[\widehat{seaITE}^i \mid X^i = x, \hat{U}^i = u \right] \quad (\text{by Assumption 5}) \\
&= \mathbb{E} \left[\widehat{seaITE}^i \mid X^i = x, \tilde{U}^i = u \right] \quad (\text{by Assumption 6: } \hat{U}^i = \tilde{U}^i) \\
&= seaMTE(u, x).
\end{aligned}$$

The second equality uses the fact that $\hat{U}^i$ is a function of (G^i, X^i) . The third equality is the first step of the proof of Proposition 1', which uses only Assumption 6. By Assumption 6, $\hat{G}^i = \psi(G^i)$ for a strictly increasing ψ , so $\hat{U}^i = 1 - F_{\hat{G}^i|X^i}(\psi(G^i))$. Therefore $\mathbb{E}[\eta_D^i(y) \mid X^i, G^i] = 0$ implies $\mathbb{E}[\eta_D^i(y) \mid X^i, \hat{U}^i] = 0$ by the law of iterated expectations. $\square$

Proposition 2 (Mean-Correct Beliefs Equate $seaMTE$ and MTE). *Let Assumptions 1, 2, 3, and zero measurement error hold. Then $\widehat{seaMTE}(u, x) = seaMTE(u, x) = MTE(u, x)$.*

Proof. With zero measurement error, $\hat{U}^i = \tilde{U}^i$ and $\widehat{seaITE}^i = seaITE^i$, so $\widehat{seaMTE}(u, x) = seaMTE(u, x)$ by definition. By the second step of the proof of Proposition 1', which uses only Assumptions 1 and 2, $\tilde{U}^i = U^i$. Then

$$\begin{aligned}
seaMTE(u, x) &= \mathbb{E} \left[\mathbb{E}^i[Y_1^i] - \mathbb{E}^i[Y_0^i] \mid \tilde{U}^i = u, X^i = x \right] \\
&= \mathbb{E} \left[\mathbb{E}[Y_1^i - Y_0^i \mid X^i, V^i] \mid U^i = u, X^i = x \right] \quad (\text{Assumption 3; } \tilde{U}^i = U^i) \\
&= \mathbb{E}[Y_1^i - Y_0^i \mid U^i = u, X^i = x] \quad (\text{law of iterated expectations}) \\
&= MTE(u, x),
\end{aligned}$$

where the law of iterated expectations applies because $U^i = F_{V^i|X^i}(V^i)$ is a function of (V^i, X^i) , so the conditioning information (U^i, X^i) is coarser than (V^i, X^i) . $\square$

Proposition 2' (Mean-Correct Beliefs with Reporting Error). *Let Assumptions 1, 2, 3, 5, and 6 hold. Then $MTE(u, x) = \widehat{seaMTE}(u, x)$.*

Proof. Under the hypotheses,

$$MTE(u, x) = seaMTE(u, x) = \widehat{seaMTE}(u, x),$$

the first equality by the proof of Proposition 2, which uses only Assumptions 1, 2, and 3 for this step, and the second by Proposition A1. $\square$

Corollary A1 (Rational Expectations with Reporting Error). *Let Assumptions 1, 5, and 6 hold. Maintain rational expectations. Then $MTE(u, x) = \widehat{seaMTE}(u, x)$.*

Proof. Rational expectations imply Assumptions 2 (with $\tilde{F}_{\mu|x} = F_{\mu|x}$) and 3, so Assumptions 1, 2, 3, 5, and 6 all hold and Proposition 2' applies. $\square$

Corollary A2 (Set Identification under Rounding). *Let Assumptions 1 and 2 hold, and suppose $\hat{G}^i = r(G^i)$ for all i for some weakly increasing function r (e.g., reports rounded to a coarse grid). Then U^i is set identified:*

$$U^i \in \left[1 - P(\hat{G} \leq \hat{G}^i | X^i), 1 - P(\hat{G} < \hat{G}^i | X^i) \right].$$

Proof. By the second step of the proof of Proposition 1', Assumptions 1 and 2 imply $F_{G|X^i}(G^i) = 1 - U^i$. Since r is weakly increasing, $\{G \leq G^i\} \subseteq \{r(G) \leq r(G^i)\}$ and $\{r(G) < r(G^i)\} \subseteq \{G < G^i\}$. Therefore

$$\begin{aligned} F_{\hat{G}|X^i}(\hat{G}^i) &= P(r(G) \leq r(G^i) | X^i) \geq P(G \leq G^i | X^i) = 1 - U^i, \\ P(\hat{G} < \hat{G}^i | X^i) &= P(r(G) < r(G^i) | X^i) \leq P(G < G^i | X^i) = P(G \leq G^i | X^i) = 1 - U^i, \end{aligned}$$

where $P(G < G^i | X^i) = P(G \leq G^i | X^i)$ because G is continuously distributed (Assumption 1). Rearranging yields $1 - F_{\hat{G}|X^i}(\hat{G}^i) \leq U^i \leq 1 - P(\hat{G} < \hat{G}^i | X^i)$. The length of the interval is $F_{\hat{G}|X^i}(\hat{G}^i) - P(\hat{G} < \hat{G}^i | X^i) = P(\hat{G} = \hat{G}^i | X^i)$. If r is strictly increasing, then $\{\hat{G} = \hat{G}^i\} = \{G = G^i\}$, which has probability zero, so the interval collapses to the point $U^i = 1 - F_{\hat{G}|X^i}(\hat{G}^i) = \hat{U}^i$. $\square$

The interval is the heap of respondents who report the same probability. In our data, ties within a heap are broken by adding a small uniform random perturbation (of order 10^{-3}) to $\hat{G}^i$ before ranking. The reported rank, and hence the cell, of 168 respondents, 150 of whom report $\hat{G}^i = 0$, depends on this perturbation.

Proposition 3 (Average Effects from Belief Levels). *Let zero measurement error and Assumption 3 hold. Then (i) $\mathbb{E}[\widehat{seaITE}^i] = ATE$; (ii) if in addition Assumption 4 holds and $\mathbb{E}[G^i] \in (0, 1)$, then $\mathbb{E}[\widehat{seaITE}^i \hat{G}^i] / \mathbb{E}[\hat{G}^i]$ equals the ex ante ATT and $\mathbb{E}[\widehat{seaITE}^i (1 - \hat{G}^i)] / \mathbb{E}[1 - \hat{G}^i]$ equals the ex ante ATU.*

Proof. Under zero measurement error, $\widehat{seaITE}^i = seaITE^i$ and $\hat{G}^i = G^i$. (i) By Assumption 3 and the law of iterated expectations,

$$\mathbb{E}[seaITE^i] = \mathbb{E}[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i]] = \mathbb{E}[Y_1^i - Y_0^i].$$

(ii) Because $\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i]$ is a function of (X^i, V^i) , the law of iterated expectations and Assumption 4 give

$$\begin{aligned}\mathbb{E}[\text{seaITE}^i G^i] &= \mathbb{E}[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] \mathbb{E}[G^i | X^i, V^i]] \\ &= \mathbb{E}[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] \mathbb{E}[D^i | X^i, V^i]] \\ &= \mathbb{E}[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] D^i],\end{aligned}$$

and $\mathbb{E}[G^i] = \mathbb{E}[\mathbb{E}[G^i | X^i, V^i]] = \mathbb{E}[D^i] = P(D^i = 1)$. Dividing,

$$\frac{\mathbb{E}[\text{seaITE}^i G^i]}{\mathbb{E}[G^i]} = \frac{\mathbb{E}[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] D^i]}{P(D^i = 1)} = \mathbb{E}\left[\mathbb{E}[Y_1^i - Y_0^i | X^i, V^i] \mid D^i = 1\right].$$

The *ATU* statement follows by the same argument with $1 - G^i$ and $1 - D^i$. $\square$

Proposition 3' (Average Effects with Reporting Error). *Let Assumptions 3 and 5 hold. Then (i) $\mathbb{E}[\widehat{\text{seaITE}}^i] = ATE$. Additionally, let Assumptions 4 and 7 hold and $\mathbb{E}[G^i] \in (0, 1)$. Then (ii) $\mathbb{E}[\widehat{\text{seaITE}}^i \hat{G}^i] / \mathbb{E}[\hat{G}^i]$ equals the ex ante *ATT* and $\mathbb{E}[\widehat{\text{seaITE}}^i (1 - \hat{G}^i)] / \mathbb{E}[1 - \hat{G}^i]$ equals the ex ante *ATU*.*

Proof. Under Assumption 5, $\widehat{\text{seaITE}}^i = \text{seaITE}^i + \bar{\eta}_1^i - \bar{\eta}_0^i$, because the common component $\bar{C}^i$ cancels in the difference, and $\mathbb{E}[\bar{\eta}_D^i | X^i, G^i] = 0$. (i) Hence $\mathbb{E}[\widehat{\text{seaITE}}^i] = \mathbb{E}[\text{seaITE}^i]$, and the proof of Proposition 3(i) applies. (ii) Under Assumption 7, $\mathbb{E}[\hat{G}^i] = \mathbb{E}[G^i]$ and

$$\mathbb{E}[\widehat{\text{seaITE}}^i \hat{G}^i] = \mathbb{E}[(\text{seaITE}^i + \bar{\eta}_1^i - \bar{\eta}_0^i)(G^i + \zeta^i)] = \mathbb{E}[\text{seaITE}^i G^i],$$

because, by Assumption 7, ζ^i is uncorrelated with seaITE^i and with each $\bar{\eta}_D^i$, and because $\mathbb{E}[\bar{\eta}_D^i G^i] = \mathbb{E}[G^i \mathbb{E}[\bar{\eta}_D^i | X^i, G^i]] = 0$. The proof of Proposition 3(ii) then applies, and the *ATU* statement follows by the same argument with $1 - \hat{G}^i$, $1 - G^i$, and $1 - D^i$. $\square$

Proposition 4 (Identification of the *PRTE* without Policy Invariance). *Let zero measurement error hold in each policy regime $p \in \{b, c\}$. Then (i) $\widehat{\text{seaPRTE}} = \text{seaPRTE}$; (ii) if Assumptions 3 and 4 hold in each regime and the average bias in the level of outcome beliefs is the same in both regimes, $\mathbb{E}[\mathbb{E}_c^i[Y_{0,c}^i] - \mathbb{E}[Y_{0,c}^i | X^i, V^i]] = \mathbb{E}[\mathbb{E}_b^i[Y_{0,b}^i] - \mathbb{E}[Y_{0,b}^i | X^i, V^i]]$, then $\mathbb{E}[G_p^i] = \mathbb{E}[D_p]$ and $\text{seaPRTE} = \text{PRTE}$; (iii) if in addition $\kappa_c = \kappa_b$, where $\kappa_p := \mathbb{E}[\text{Cov}(D_p, Y_{1,p} - Y_{0,p} | X, V)]$, then $\text{PRTE} = \frac{\mathbb{E}[Y_c] - \mathbb{E}[Y_b]}{\mathbb{E}[D_c] - \mathbb{E}[D_b]}$; in particular, if $\mathbb{E}[Y_{D,p} | D_p, X, V] = \mathbb{E}[Y_{D,p} | X, V]$ for $D \in \{0, 1\}$, then $\kappa_p = 0$ and $\bar{Y}_p = \mathbb{E}[Y_p]$ in each regime.*

Proof. (i) Under zero measurement error the reports equal the beliefs, so $\hat{Y}_p^{\text{sea}} = \bar{Y}_p^{\text{sea}}$ and $\mathbb{E}[\hat{G}_p^i] = \mathbb{E}[G_p^i]$ in each regime, and the two ratios coincide. (ii) By Assumption 4 and the law of iterated expectations, $\mathbb{E}[G_p^i] = \mathbb{E}[\mathbb{E}[G_p^i | X, V]] = \mathbb{E}[P(D_p = 1 | X, V)] = \mathbb{E}[D_p]$.

Hence the denominators in equation (14) coincide. For the numerators, write $seaITE_p^i := \mathbb{E}_p^i[Y_{1,p}^i] - \mathbb{E}_p^i[Y_{0,p}^i]$ and rearrange equations (12) and (13) as

$$\begin{aligned}\bar{Y}_p^{sea} &= \mathbb{E}[\mathbb{E}_p^i[Y_{0,p}^i]] + \mathbb{E}[G_p^i seaITE_p^i], \\ \bar{Y}_p &= \mathbb{E}[\mathbb{E}[Y_{0,p}^i|X^i, V^i]] + \mathbb{E}[P(D_p = 1|X^i, V^i) \mathbb{E}[Y_{1,p}^i - Y_{0,p}^i|X^i, V^i]].\end{aligned}$$

By Assumption 3, $seaITE_p^i = \mathbb{E}[Y_{1,p}^i - Y_{0,p}^i|X^i, V^i]$ is a function of (X^i, V^i) , so the law of iterated expectations and Assumption 4 give $\mathbb{E}[G_p^i seaITE_p^i] = \mathbb{E}[\mathbb{E}[G_p^i|X^i, V^i] seaITE_p^i] = \mathbb{E}[P(D_p = 1|X^i, V^i) \mathbb{E}[Y_{1,p}^i - Y_{0,p}^i|X^i, V^i]]$, as in the proof of Proposition 3(ii). Subtracting, $\bar{Y}_p^{sea} - \bar{Y}_p = \mathbb{E}[\mathbb{E}_p^i[Y_{0,p}^i] - \mathbb{E}[Y_{0,p}^i|X^i, V^i]]$, the average bias in the level of outcome beliefs, which by hypothesis is the same in both regimes and therefore cancels in $\bar{Y}_c^{sea} - \bar{Y}_b^{sea}$. Hence the numerators coincide as well. Each regime's objects are elicited within that regime; the equal-bias hypothesis is the only restriction across regimes. (iii) By the law of iterated expectations and the definition of a conditional covariance,

$$\begin{aligned}\mathbb{E}[D_p Y_{1,p}] &= \mathbb{E}[P(D_p = 1|X, V) \mathbb{E}[Y_{1,p}|X, V]] + \mathbb{E}[\text{Cov}(D_p, Y_{1,p} | X, V)], \\ \mathbb{E}[(1 - D_p) Y_{0,p}] &= \mathbb{E}[P(D_p = 0|X, V) \mathbb{E}[Y_{0,p}|X, V]] - \mathbb{E}[\text{Cov}(D_p, Y_{0,p} | X, V)].\end{aligned}$$

Summing gives $\mathbb{E}[Y_p] = \bar{Y}_p + \kappa_p$. Hence $\mathbb{E}[Y_c] - \mathbb{E}[Y_b] = \bar{Y}_c - \bar{Y}_b + (\kappa_c - \kappa_b)$, and with $\kappa_c = \kappa_b$ and $\mathbb{E}[D_p]$ in the denominator the $PRTE$ equals the realized ratio. Under mean independence, $\mathbb{E}[Y_{D,p}|D_p, X, V] = \mathbb{E}[Y_{D,p}|X, V]$ implies $\text{Cov}(D_p, Y_{D,p} | X, V) = 0$ for each D , so $\kappa_p = 0$ and $\mathbb{E}[Y_p] = \bar{Y}_p$. $\square$

Proposition 4' (*seaPRTE with Reporting Error*). *Let Assumptions 5 and 7 hold in each policy regime $p \in \{b, c\}$, with C^i the same in both regimes. Then $\widehat{seaPRTE} = seaPRTE$.*

Proof. Fix a regime p and write $seaITE_p^i := \mathbb{E}_p^i[Y_{1,p}^i] - \mathbb{E}_p^i[Y_{0,p}^i]$. Rearranging equation (12),

$$\bar{Y}_p^{sea} = \mathbb{E}[\mathbb{E}_p^i[Y_{0,p}^i]] + \mathbb{E}[G_p^i seaITE_p^i],$$

and likewise $\hat{\bar{Y}}_p^{sea} = \mathbb{E}[\hat{\mathbb{E}}_p^i[Y_{0,p}^i]] + \mathbb{E}[\hat{G}_p^i \widehat{seaITE}_p^i]$. By Assumption 7, $\mathbb{E}[\hat{G}_p^i] = \mathbb{E}[G_p^i]$, so the denominators of $\widehat{seaPRTE}$ and $seaPRTE$ coincide. By Assumption 5, $\mathbb{E}[\hat{\mathbb{E}}_p^i[Y_{0,p}^i]] = \mathbb{E}[\mathbb{E}_p^i[Y_{0,p}^i]] + \mathbb{E}[\bar{C}^i]$, and the argument in the proof of Proposition 3'(ii) gives $\mathbb{E}[\hat{G}_p^i \widehat{seaITE}_p^i] = \mathbb{E}[G_p^i seaITE_p^i]$. Hence $\hat{\bar{Y}}_p^{sea} = \bar{Y}_p^{sea} + \mathbb{E}[\bar{C}^i]$, and because C^i is the same in both regimes, $\mathbb{E}[\bar{C}^i]$ cancels in $\hat{\bar{Y}}_c^{sea} - \hat{\bar{Y}}_b^{sea}$, so the numerators coincide as well. $\square$

Proposition A2 (*MTE with Random Error in Ranks*). *Let Assumptions 1 and 3 hold, let $\hat{G}$ be continuously distributed conditional on X , and let Assumption 5 hold with $\hat{U}^i$ in place of G^i , that is, $\mathbb{E}[\eta_D^i(y) | X^i, \hat{U}^i] = 0$ for all y and D . Then:*

- (i) $\widehat{seaMTE}(b, x) = \mathbb{E} \left[MTE(U^i, x) \mid \hat{U}^i = b, X^i = x \right];$
- (ii) $\int_0^1 \widehat{seaMTE}(b, x) db = \int_0^1 MTE(u, x) du;$
- (iii) $Var \left[\widehat{seaMTE}(\hat{U}^i, x) \mid X^i = x \right] \leq Var[MTE(U^i, x) \mid X^i = x]$ and $\inf_u MTE(u, x) \leq \widehat{seaMTE}(b, x) \leq \sup_u MTE(u, x);$
- (iv) *if in addition the conditional distribution of U^i given $(\hat{U}^i = b, X^i = x)$ is stochastically increasing in b and $MTE(\cdot, x)$ is weakly monotone, then $\widehat{seaMTE}(\cdot, x)$ is weakly monotone in the same direction.*

Recall that continuity of V and $\hat{G}$ conditional on X makes both U^i and $\hat{U}^i$ uniformly distributed on $[0, 1]$ conditional on X .

Proof. Under the outcome-report condition, $\widehat{seaITE}^i = seaITE^i + \bar{\eta}_1^i - \bar{\eta}_0^i$ with $\mathbb{E}[\bar{\eta}_D^i \mid X^i, \hat{U}^i] = 0$, because the common component $\bar{C}^i$ cancels in the difference (as in the proof of Proposition A1), so $\widehat{seaMTE}(b, x) = \mathbb{E}[seaITE^i \mid \hat{U}^i = b, X^i = x]$. Under Assumption 3, $seaITE^i = \mathbb{E}[Y_1^i - Y_0^i \mid X^i, V^i]$, and conditional on (U^i, X^i) , V^i is the U^i -quantile of $F_{V \mid X^i}$ for almost every U^i (Assumption 1), so $seaITE^i$ is almost surely a function of (U^i, X^i) and equals $MTE(U^i, X^i)$. (i) Hence $\mathbb{E}[seaITE^i \mid \hat{U}^i = b, X^i = x] = \mathbb{E}[MTE(U^i, x) \mid \hat{U}^i = b, X^i = x]$. (ii) Both sides equal $\mathbb{E}[seaITE^i \mid X^i = x]$ by the law of iterated expectations, using that U and $\hat{U}$ are each uniform on $[0, 1]$ given X . (iii) By (i), $\widehat{seaMTE}(\hat{U}, x) = \mathbb{E}[MTE(U, x) \mid \hat{U}, X = x]$, and the variance of a conditional expectation is weakly smaller than the variance of the variable itself (law of total variance); the bounds follow because a mixture of values of $MTE(\cdot, x)$ lies within its range. (iv) A distribution that increases in the first-order stochastic dominance order raises (lowers) the expectation of any weakly increasing (decreasing) function, so $b \mapsto \mathbb{E}[MTE(U, x) \mid \hat{U} = b, X = x]$ inherits the direction of monotonicity of $MTE(\cdot, x)$. $\square$

Proposition A2 is Proposition 2' without the rank conditions. Part (i) is the statement in Section 2.5 that the measured curve is a flattened version of the true curve, and parts (ii), (iii), and (iv) are its three consequences: the ATE is preserved, the heterogeneity of the measured curve bounds the true heterogeneity from below, and weak monotonicity is inherited when the measured rank is a monotone signal of the latent rank. When the measured rank equals the latent rank, as under Assumptions 2 and 6, the conditional distribution in part (i) is a point mass, and the proposition reduces to Proposition 2'.

Under Assumption 6 the measured rank is a function of (X, G) and $\hat{G}$ is continuous whenever G is, so the hypotheses of Proposition 2' imply those of Proposition A2. Without Assumption 3, the same statements hold with the subjective gain at the latent rank, $\mathbb{E}[seaITE^i \mid U^i = u, X^i = x]$, in place of $MTE(u, x)$, provided that, given the latent rank, expected gains do

not depend on the measured rank, $\mathbb{E}[seaITE^i | U^i, \hat{U}^i, X^i] = \mathbb{E}[seaITE^i | U^i, X^i]$. The proof is the same, with this condition replacing the step that made $seaITE^i$ a function of (U^i, X^i) . Under Assumption 3 that condition holds automatically, which is why Proposition A2 does not state it.

B Extensions: Multi-Dimensional Shocks and a General Formulation

The model in Section 2.1 specifies scalar shocks and a structural outcome function. This is convenient for exposition, but none of the identification results depend on it. This appendix makes that claim precise. Appendix B.1 extends the model to multi-dimensional shocks, Appendix B.2 restates the environment as a general potential-outcomes model that nests the structural model in the body of the paper, and Appendix B.3 extends the results to general functionals of the treatment-contingent outcome distributions and to general outcome spaces. Throughout, all referenced expectations and integrals are assumed finite.

B.1 Multi-Dimensional Shocks

Let the selection shock $\mu \in \mathbb{R}^{d_\mu}$ and the outcome shock $\epsilon \in \mathbb{R}^{d_\epsilon}$ be random vectors drawn, as in Section 2.1, at $t = 1$ and $t = 2$ from conditional distributions $F_{\mu|X}$ and $F_{\epsilon|X}$, with $(\mu, \epsilon) \perp V|X$.¹⁹

Because treatment is binary and selection is a threshold rule in the scalar latent type V , the selection shock must enter the treatment decision through a scalar index. Let m be a measurable function mapping (X^i, μ^i) into the selection index $\nu^i := m(X^i, \mu^i) \in \mathbb{R}$, and replace selection equation (1) by

$$D^i = \begin{cases} 1 & \text{if } \nu^i \geq V^i \\ 0 & \text{if } \nu^i < V^i, \end{cases} \quad (21)$$

and let $F_{\nu|X}$ denote the CDF of ν conditional on X induced by $F_{\mu|X}$ and m . Outcomes may depend on the full shock vectors, not only on the index: $Y_D^i(X^i, V^i, \mu^i, \epsilon^i)$ with vector-valued μ^i and ϵ^i .

The objects of Section 2 change only by reinterpretation. Under rational expectations, equation (2) becomes $G^i = 1 - F_{\nu|X^i}(V^i)$: individuals forecast the scalar index, however many primitive shocks lie behind it. Assumption 2 is restated with a common subjective CDF $\tilde{F}_{\nu|x}$ of the *index* in place of $\tilde{F}_{\mu|x}$. Individuals need not hold common, or even well-formed, beliefs about the full vector μ , only a common monotone map from V into the probability of treatment. Assumptions 1, 5, and 6 and all definitions (U , $\tilde{U}$, $\hat{U}$, MTE , $seaITE$, $seaMTE$, and their measured counterparts) are unchanged, except that the integral representation of

¹⁹Independence between μ and ϵ conditional on X , which the body of the paper also assumes, is not needed for any identification result. It is used only to write the MTE integral in equation (7) in iterated form and in the ex ante/ex post discussion.

the *MTE* integrates over the joint conditional law $F_{\mu, \epsilon|x}$ of the shocks:

$$MTE(u, x) = \int [Y_1^i(x, v_u, m', e) - Y_0^i(x, v_u, m', e)] dF_{\mu, \epsilon|x}(m', e),$$

which factors into the iterated integral in equation (7) when $\mu \perp \epsilon|X$.

Proposition B1 (Multi-Dimensional Shocks). *In the model with multi-dimensional shocks, Propositions 1, 1', A1, 2, 2', 3, 3', 4, 4', and A2 and Corollaries A1 and A2 hold verbatim, with the distributions of the selection index ν in place of the corresponding distributions of μ .*

Proof. Assumptions 1, 2, 5, 6, and 7 restrict only the joint conditional distribution of $(V, G, \hat{G})$ and $(h_0, h_1, \hat{h}_0, \hat{h}_1)$ given X , and the proofs in Appendix A operate exclusively on that joint distribution: the shocks appear in no proof. Assumptions 3 and 4 and part (iii) of Proposition 4 reference the conditional expectations $\mathbb{E}[Y_1^i - Y_0^i|X^i, V^i]$ and $P(D^i = 1|X^i, V^i)$ and the conditional covariance κ_p , which are well defined whatever the dimension of the shocks that generate D^i and Y_D^i . Hence no result depends on d_μ or d_ϵ . $\square$

The multi-dimensional case separates selection-relevant from outcome-only variation. A component of μ may enter the selection index but not outcomes, or outcomes but not the index. In the childbirth application, the promotion is a $t = 1$ event that plausibly enters both, while a change in the workplace environment may affect the taste for work after childbirth without affecting the fertility decision. This separation refines the ex ante/ex post remark in Section 2.2. Within a (U, X) cell, treatment status is determined by the index alone ($D^i = 1$ if and only if $\nu^i \geq v_u$), so the ex post treated–untreated difference in equation (9) recovers $MTE(u, x)$ whenever potential outcomes are mean-independent of the index conditional on (X, V) , i.e., $\mathbb{E}[Y_D^i|\nu^i, X^i, V^i] = \mathbb{E}[Y_D^i|X^i, V^i]$. Outcomes may depend arbitrarily on shock components that are conditionally independent of the index without invalidating that difference.

B.2 A General Potential-Outcomes Formulation

The structural model in the body of the paper is one data generating process consistent with our identification results. The results themselves hold in a more general environment specified directly by the joint distribution of a small set of random elements.

Primitives. Each individual i is described by a draw of the random element

$$(X, V, \nu, Y_0, Y_1, G, h_0, h_1, \hat{G}, \hat{h}_0, \hat{h}_1),$$

where $X \in \mathbb{R}^{d_x}$ are observables; $V \in \mathbb{R}$ is a scalar latent type known to the individual at $t = 0$; $\nu \in \mathbb{R}$ is a scalar selection index not known at $t = 0$; treatment is the threshold rule $D = \mathbb{1}\{\nu \geq V\}$; (Y_0, Y_1) are potential outcomes, with realized outcome $Y = (1 - D)Y_0 + DY_1$; (G, h_0, h_1) are the individual's subjective beliefs at $t = 0$, with G the subjective probability that $D = 1$ and h_D the subjective density of Y_D ; and $(\hat{G}, \hat{h}_0, \hat{h}_1)$ are the survey reports. Densities are taken with respect to a fixed σ -finite dominating measure λ on the outcome space, and all integrals over y are integrals against λ . For continuous outcomes λ is Lebesgue measure, and for discrete outcomes, as in our application where the outcome is binary, it is the counting measure.

Three features of this formulation deserve emphasis. First, the joint distribution of (Y_0, Y_1, ν) given (X, V) is unrestricted; in particular, the selection index may be arbitrarily dependent on the potential outcomes. This is the formal statement of the point developed in Section 2.6 that our approach requires no exclusion restriction. No variable in the model needs to shift selection while being excluded from outcomes. Second, the substantive structural restriction is the threshold-crossing form $D = \mathbb{1}\{\nu \geq V\}$ with a scalar V that is known to the individual when beliefs are measured. This is the belief-side analog of the nonparametric selection structure that Vytlacil (2002) shows is equivalent to the monotonicity condition of Imbens and Angrist (1994). Third, V must be interpreted as exhaustive: because Assumption 2 requires individuals with the same (X, V) to hold the same treatment belief, V captures all $t = 0$ private heterogeneity that is relevant for forecasting treatment. The framework thus requires forecast-relevant private heterogeneity to be effectively one-dimensional conditional on observables.

Assumptions and Results. Assumptions 1, 5, and 6 apply verbatim, since they restrict only the joint distribution of beliefs, reports, and V given X . Assumption 2 takes the marginally more general form: for each x there exists a weakly decreasing function φ_x , common to all individuals with $X^i = x$, such that $G^i = \varphi_{X^i}(V^i)$. (The CDF form in the body, $\varphi_x = 1 - \tilde{F}_{\nu|x}$, is a convenient special case that nests rational expectations; the proofs use only monotonicity.) Take U , $\tilde{U}$, and $\hat{U}$ as defined in the body; by Assumption 1, U is uniformly distributed on $[0, 1]$ conditional on X , and selection can equivalently be written $D = \mathbb{1}\{F_{V|X}(\nu) \geq U\}$. The treatment-effect objects are defined exactly as before: $MTE(u, x) := \mathbb{E}[Y_1 - Y_0 | U = u, X = x]$, $seaITE := \int y [h_1(y) - h_0(y)] d\lambda(y)$, $seaMTE(u, x) := \mathbb{E}[seaITE | \tilde{U} = u, X = x]$, and the measured counterparts using $(\hat{G}, \hat{h}_D)$ in place of (G, h_D) .

Proposition B2 (General Formulation). *In the general formulation: (i) under Assumptions 1, 2, and 6, $U^i = \tilde{U}^i = \hat{U}^i$; (ii) under Assumptions 1, 2, 5, and 6, $\widehat{seaMTE}(u, x) = seaMTE(u, x)$; and (iii) under Assumptions 1, 2, and 3, $seaMTE(u, x) = MTE(u, x)$.*

Proof. The proofs of Propositions 1' and A1 and the subjective-to-objective step of the proof of Proposition 2 in Appendix A apply verbatim, reading the weakly decreasing map φ_x wherever $1 - \tilde{F}_{\mu|x}$ appears. They use only Assumptions 1, 2, 3, 5, and 6, the definitions above, and, for (iii), the law of iterated expectations; none of these reference a structural outcome function or the shocks. $\square$

Rational Expectations and Aggregation. In the general model, the individual's information at $t = 0$ is (X, V) , so rational expectations means $G = P(D = 1|X, V)$. If, in addition, the selection index satisfies the conditional independence restriction $\nu \perp V|X$, then $G = P(\nu \geq V|X, V) = 1 - F_{\nu|X}(V)$ almost surely, so Assumption 2 holds with $\varphi_x = 1 - F_{\nu|x}$. In this formulation, the single condition $\nu \perp V|X$ reconciles rational expectations with the common forecasting rule. The structural model of Section 2.1 is the special case $\nu = m(X, \mu)$ and $Y_D = Y_D(X, V, \mu, \epsilon)$ with $(\mu, \epsilon) \perp V|X$, which implies $\nu \perp V|X$. Under rational expectations, two aggregation identities follow from the law of iterated expectations:

$$P(D^i = 1|X^i, V^i) = G^i \quad \text{and} \quad P(D^i = 1|X^i) = \mathbb{E}[G^i|X^i].$$

The individual's belief is itself a propensity score—the forecast of treatment given the information (X, V) —while the econometrician's propensity score given X is the average of individual beliefs. The second identity implies that, under rational expectations and a stable environment, average reported treatment probabilities should match average treatment rates conditional on observables. This is the comparison implemented in Section 3.4, where we compare stated fertility expectations with realized fertility in the registry data. Because that comparison uses reported probabilities $\hat{G}$, it also tests whether reported treatment probabilities equal beliefs on average.

B.3 General Functionals of Outcome Distributions

Because the survey elicits the entire subjective distribution of treatment-contingent outcomes, the theory is not limited to means. Let g be a measurable real-valued function on the outcome space (with the required absolute integrability), and define

$$\begin{aligned} MTE_g(u, x) &:= \mathbb{E}[g(Y_1^i) - g(Y_0^i) | U^i = u, X^i = x], \\ seaITE_g^i &:= \int g(y) [h_1^i(y) - h_0^i(y)] d\lambda(y) = \mathbb{E}^i[g(Y_1^i)] - \mathbb{E}^i[g(Y_0^i)], \\ seaMTE_g(u, x) &:= \mathbb{E}[seaITE_g^i | \tilde{U}^i = u, X^i = x], \end{aligned}$$

with measured counterparts $\widehat{seaITE}_g^i$ and $\widehat{seaMTE}_g(u, x)$ defined using $\hat{h}_D^i$ and $\hat{U}^i$. Setting $g(y) = y$ recovers the objects in the body of the paper; $g(y) = \mathbb{1}\{y \leq t\}$ yields distributional marginal treatment effects, $P(Y_1^i \leq t | U^i = u, X^i = x) - P(Y_0^i \leq t | U^i = u, X^i = x)$; and $g(y) = y^k$ yields higher-moment effects.

Proposition B3 (General Functionals). *Fix any such g .*

(i) *Under Assumptions 1, 2, 5, and 6, $\widehat{seaMTE}_g(u, x) = seaMTE_g(u, x)$.*

(ii) *If Assumptions 1 and 2 hold and*

$$\mathbb{E}^i [g(Y_1^i)] - \mathbb{E}^i [g(Y_0^i)] = \mathbb{E} [g(Y_1^i) - g(Y_0^i) | X^i, V^i] \quad \text{for all } i,$$

then $seaMTE_g(u, x) = MTE_g(u, x)$.

Proof. Replace y by $g(y)$ throughout the proof of Proposition A1 and the subjective-to-objective step of the proof of Proposition 2 in Appendix A. Because Assumption 5 restricts reported beliefs pointwise in y , the common error $C^i(y)$ cancels within individual across treatment states under any g , and

$$\int g(y) \mathbb{E} [\eta_1^i(y) - \eta_0^i(y) | X^i, \hat{U}^i] d\lambda(y) = 0$$

by the conditional mean-zero property in Assumption 5 and Fubini's theorem. $\square$

Three remarks conclude. First, a single measurement-error assumption delivers part (i) for all g simultaneously, because Assumption 5 restricts the reported densities pointwise in y . This robustness to reporting error applies to functionals that are linear in the belief densities. Nonlinear functionals, for example quantile treatment effects obtained by inverting the identified treatment-contingent outcome distributions, are identified under zero measurement error, when the belief distributions themselves are observed. Second, nothing requires the outcome space to be the real line: outcomes may be discrete, multi-dimensional, or mixed, with h_D a density with respect to a dominating measure on that space and g any integrable real-valued function. In this sense the restriction $Y_D \in \mathbb{R}$ in the body of the paper is also expositional. Third, when the outcome is binary, as in our application, the subjective mean determines the entire belief distribution, so every functional of it is a function of the mean and the mean-based exposition in the body of the paper is without loss of generality.

C Interpreting the State-Contingent Survey Questions

A scenario prompt (“suppose you do have a child”) admits two readings. Under the interventional reading, maintained throughout the paper, the respondent imagines her treatment state set to D while holding fixed her information about the shocks, and she reports $\mathbb{E}^i[Y_D^i]$. Under the observational reading, she treats the supposition as information about how her future unfolds and reports $\mathbb{E}^i[Y_D^i | D^i = D]$. Suppose her beliefs about the selection shock μ^i and the potential outcomes (Y_0^i, Y_1^i) form one joint distribution P^i , under which treatment is the event $\mu^i \geq V^i$ because she knows V^i and the selection rule. Conditioning on $D^i = D$ then restricts μ^i to one side of V^i and does nothing else, so the two readings deliver the same response if and only if

$$\mathbb{E}^i[Y_D^i | D^i = D] = \mathbb{E}^i[Y_D^i] \quad \text{for } D \in \{0, 1\} \text{ and all } i. \quad (22)$$

Condition (22) holds whenever outcome beliefs do not depend on the selection shock, as when the outcome function takes the form $Y_D^i(X^i, V^i, \epsilon^i)$ of Section 2.2. Only dependence of outcome beliefs on the selection shock matters. As in Appendix B.1, shocks that affect outcomes but are unrelated to the fertility decision leave condition (22) intact, because conditioning on the birth state is uninformative about them. When condition (22) fails, the observational contrast $\mathbb{E}^i[Y_1^i | D^i = 1] - \mathbb{E}^i[Y_0^i | D^i = 0]$ is the subjective, individual-level analog of the ex post treated–untreated difference in equation (9). It combines the causal effect of treatment with the information that treatment status conveys about μ^i .

The issue is confined to the outcome questions. The treatment probability G^i is a forecast rather than a causal object. The nested suppositions used to elicit it, the probability of trying to conceive and the probabilities of a birth conditional on trying and on not trying, are averaged with elicited weights, so the law of total probability recovers the marginal probability of a birth under either reading. The partner-leave nesting within the birth state is averaged over in the same way. The identification of U^i from treatment beliefs is therefore unaffected by the reading. So is the selection problem that motivates conditioning on the latent rank, because condition (22) restricts only the dependence of outcome beliefs on the within-type shock μ^i , and outcome beliefs may depend arbitrarily on the latent type V^i . Under the observational reading, moreover, weighting each respondent’s treatment-conditional reports by her treatment probability recovers the regime mean $\mathbb{E}[Y_p]$ by the law of iterated expectations, so the *PRTE* of Section 2.4 is identified under either reading.

The comparison of survey-based and event-study *ATTs* in Section 3.4.2 tests condition (22) in the child state. Both series are child-contingent paths normalized at the pre-birth month, and the event-study path targets $\mathbb{E}[Y_1^i | D^i = 1]$ at each horizon. Under the observational

reading, with Assumption 4 and treatment-conditional beliefs correct in the mean, the survey path recovers that target because weighting the treatment-conditional expectation by G^i recovers the treated mean by the law of iterated expectations:

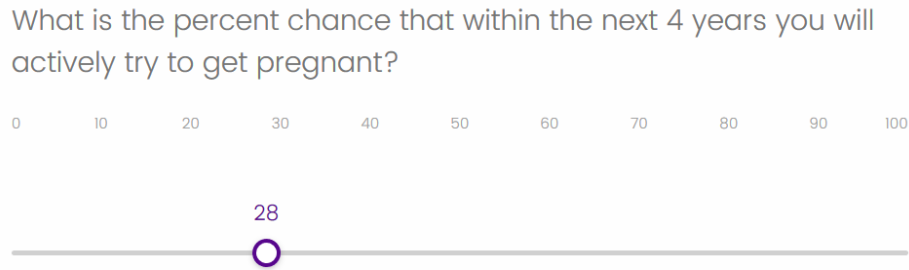
$$\frac{\mathbb{E}[G^i \mathbb{E}[Y_1^i | D^i = 1, X^i, V^i]]}{\mathbb{E}[G^i]} = \frac{\mathbb{E}[\mathbb{E}[D^i Y_1^i | X^i, V^i]]}{P(D^i = 1)} = \mathbb{E}[Y_1^i | D^i = 1] .$$

Under the interventional reading, the survey path is the G -weighted average of $\mathbb{E}^i[Y_1^i]$, which coincides with the target whenever beliefs about Y_1 do not depend on μ . The close alignment in Figure 5 therefore shows that the two readings deliver the same child-state responses in our data.

D Fertility and Labor Supply Questions and the Response Slider

Respondents answer every probability question on the slider shown in Figure A1. The slider starts at a random position, and a screen shown before the first question explains how it works. As the respondent moves the slider, the selected percent chance is displayed, so she sees the number she is reporting.

Figure A1: Response Slider.



The probability of having a child is elicited in three questions, shown sequentially:

- What is the percent chance that, within the next 4 years, you will actively try to get pregnant?
- Suppose that, within the next 4 years, you **do** actively try to get pregnant. What is the percent chance that you will have a child in the next 4 years?
- Suppose that, within the next 4 years, you **do not** actively try to get pregnant. What is the percent chance that you will have a child in the next 4 years?

The reported treatment probability, $\hat{G}^i$, compounds the three responses as in Section 3.2.1.

The probability of working with a child is elicited in three questions. The first asks for the probability that the partner takes parental leave. The second and third ask for the probability of working at each horizon conditional on the partner taking and not taking leave, in a set of stacked sliders with one line per horizon:

- Suppose that you do have a child in the next 4 years. What is the percent chance that your partner will take parental leave of at least two months in the first year after the child is born?

- Suppose you do have a child in the next 4 years and your partner **does** take at least two months parental leave. In this scenario, what is the percent chance you will be working exactly $[[12/3]$ months before/ $[3/9/18/36]$ months after] your child is born?
- Suppose you do have a child in the next 4 years and your partner **does not** take at least two months parental leave. In this scenario, what is the percent chance you will be working exactly $[[12/3]$ months before/ $[3/9/18/36]$ months after] your child is born?

The child-contingent work expectation at each horizon, $\widehat{\mathbb{E}}^i [Y_{1,s}^i]$, compounds the two branches with the reported leave probability as in equation (18).

E Balance Test for Danish Survey

Table A1: **Summary statistics for participants and nonparticipants**

Variable	(1) Non-participants	(2) Participants	(3) Difference
Age	28.625 (6.606)	29.413 (6.451)	0.788*** (0.069)
Partner	0.518 (0.500)	0.606 (0.489)	0.088*** (0.005)
Number of Children	1.247 (1.197)	1.408 (1.131)	0.161*** (0.012)
Had baby in 2017	0.068 (0.252)	0.082 (0.274)	0.014*** (0.003)
Non-DK citizenship	0.197 (0.397)	0.095 (0.294)	-0.101*** (0.003)
Income (1000DKK)	175.549 (186.734)	227.839 (193.508)	52.289*** (2.051)
Liquid assets (1000DKK)	75.029 (484.208)	87.185 (206.216)	12.156*** (3.119)
House owner	0.237 (0.425)	0.338 (0.473)	0.101*** (0.005)
Student	0.303 (0.460)	0.309 (0.462)	0.006 (0.005)
Educ basic	0.220 (0.415)	0.126 (0.332)	-0.094*** (0.004)
Educ high school	0.209 (0.407)	0.186 (0.389)	-0.023*** (0.004)
Educ vocational	0.210 (0.407)	0.207 (0.405)	-0.003 (0.004)
Educ middle	0.120 (0.325)	0.189 (0.391)	0.069*** (0.004)
Educ College	0.149 (0.356)	0.231 (0.421)	0.082*** (0.004)
Observations	39,696	11,304	51,000

Notes: Characteristics are obtained from administrative registries and are measured in 2018. Standard deviations in parentheses in columns (1) and (2). Standard errors in parentheses in column (3). * 10%, ** 5%, *** 1% level of significance.

F Estimating Average Treatment Effect on the Treated (ATT s): Subjective Expectations Method compared to Historical Event Study

In Section 3.4.2 we estimate the ATT using an event-study regression. In practice, we separately estimate ATT s for women giving birth to their first, second, and third child ($j = 1, 2, 3$) during 2010m1-2015m12. In each of these estimations we control for the age of the mother and for the timing of adjacent children that the mother has given birth to.

Specifically, we estimate the ATT k periods from childbirth for women giving birth to their child number j , denoted ATT_k^j , as the event-study coefficient δ_k^j in the following regression:

$$y_{it}^j = \sum_{\substack{k=-24 \\ k \neq -9}}^{36} D_{it}^{j,k} \delta_k^j + \gamma^j X_{it} + u_{it}^j \quad (23)$$

where y_{it} is a dummy variable taking the value one if individual i is working in month t . D_{it}^k is a dummy taking the value one if i gave birth to a child k months ago, and X is a vector of covariates including dummies for the age of the mother and event dummies for adjacent children. Including dummies for the timing of adjacent children is required to be able to estimate the direct effect of the child under consideration. For example, women who get a second child tend to have the second child around two years after the first child, and estimating the effect of the first child without controlling for the timing of the second child would confound the effect of the first and the second child.

To arrive at an overall ATT estimate, we average the estimates using frequency weights.

$$ATT_k = \sum_j P(j) \times ATT_k^j$$

where $P(j)$ is the sample share of mothers giving birth to their j th child.

F.1 ATT Estimates Based on the Sun and Abraham (2021) Estimator

The estimator proposed by Sun and Abraham (2021) first estimates treatment-cohort-specific average treatment effects on the treated and then weights them together using frequency weights. In our context, a treatment cohort consists of women giving birth to a child in a particular year $\times$ month cell. We include women giving birth to children during the period 2010m1-2015m12, i.e., the data include 72 treatment cohorts.

Specifically, the cohort-specific ATT pertaining to k periods from childbirth for cohort e , denoted $CATT_{e,k}$, is estimated as the cohort-specific event-study coefficient δ_k^e in the following regression:

$$y_{it} = \sum_{e=2010m1}^{2015m12} \sum_{\substack{k=-24 \\ k \neq -9}}^{36} D_{it}^{e,k} \delta_k^e + \gamma X_{it} + u_{it} \quad (24)$$

where y_{it} is a dummy variable taking the value one if individual i belonging to treatment cohort e is working in month t . $D_{it}^{e,k}$ is a treatment-cohort specific dummy taking the value 1 in period k after childbirth, and X is a vector of covariates including dummies for the age of the mother and event dummies for adjacent children.

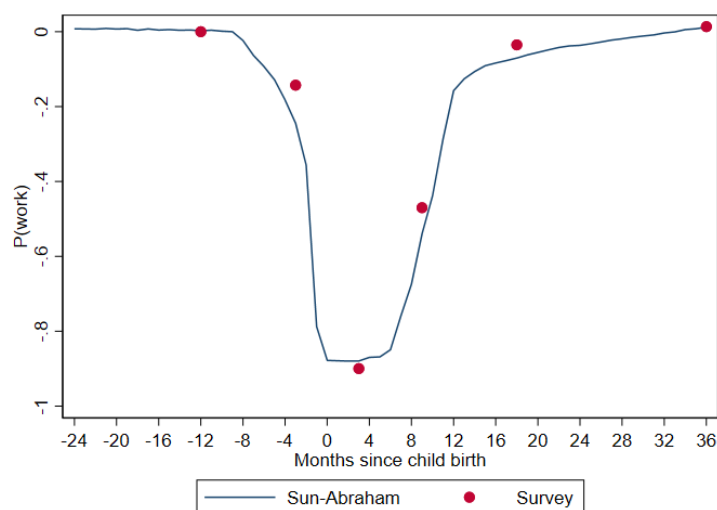
The average treatment effect k periods from childbirth, ATT_k , is calculated as the frequency-weighted average of the cohort-specific effects $CATT_{e,k}$:

$$ATT_k = \sum_e P(e) \times CATT_{e,k}$$

where $P(e)$ is the sample share of cohort e .

Implementing the Sun-Abraham estimator requires estimating many parameters. There are $K = 61$ event periods ($k = -24, \dots, 36$) for each of the $E = 72$ treatment cohorts ($e = 2010m1, \dots, 2015m12$), so we estimate $(K - 1)E = 4,320$ parameters. Following Appendix F, we estimate these parameters separately for each child number ($j = 1, 2, 3$). In our data there are too few mothers who gave birth to a third child to be able to estimate $CATT_{e,k}$ for this group. We therefore include only observations for mothers giving birth to child number 1 and 2. Figure A2 shows the result. For all practical purposes, the figure is identical to Figure 5.

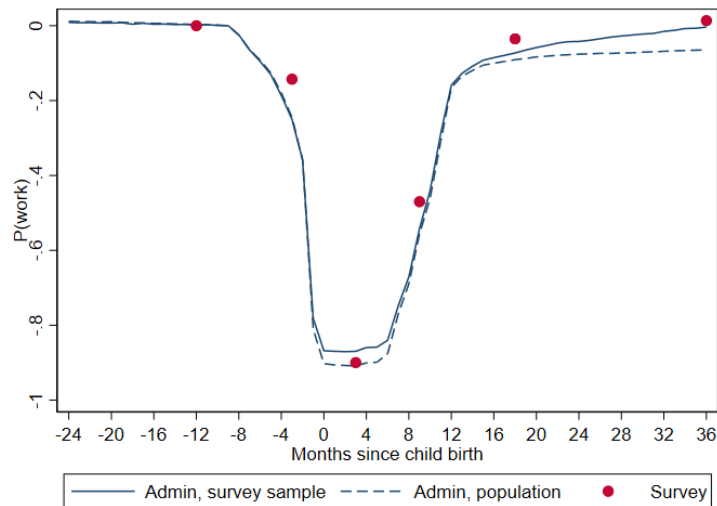
Figure A2: Estimating Average Treatment Effect on the Treated (ATT 's): Subjective Expectations Method compared to Historical Event Study Using the Sun and Abraham (2021) Estimator



Notes: The figure shows the percentage of women working at different horizons relative to childbirth, where the fraction of women working in period -12 is set to zero. Estimates thus give the deviation in labor supply from the level at month -12. The red dots show the survey ATT , the average across respondents of the $\widehat{seATE}_s^i$ of Section 3.2.1, weighting by the relative probability of having a child. The solid blue line shows event study ATT estimates using the Sun and Abraham (2021) estimator. The sample includes women who are recorded in the administrative registries to have given birth during the period 2010–2015 and who participate in the survey.

F.2 Survey Sample versus Population

Figure A3: Estimating Average Treatment Effect on the Treated (ATT s): Subjective Expectations Method compared to Historical Event Study



Notes: The figure shows the percentage of women working at different horizons relative to childbirth, where the fraction of women working in period -12 is set to zero. Estimates thus give the deviation in labor supply from the level at month -12. The red dots show the survey ATT , the average across respondents of the $\widehat{seaITE}_s^i$ of Section 3.2.1, weighting by the relative probability of having a child. The dashed blue line shows event study ATT estimates, i.e., δ from equation (20), based on registry data for the entire Danish population of women aged 20–40 who had children during the period 2010–2015. The solid line shows event study estimates based on childbirths recorded in the administrative registries for the women who are in the survey sample.

G Correlation of $\widehat{seaITEs}$ and Covariates (Baseline Scenario)

Table A2: How $\widehat{seaITEs}$ correlate with Covariates in the Baseline Scenario

	(1)	(2)	(3)	(4)
	Month 3	Month 9	Month 18	Month 36
1 child	-0.043*** (0.008)	-0.057*** (0.011)	0.003 (0.007)	0.009 (0.007)
2 children	-0.025* (0.013)	-0.042** (0.017)	0.009 (0.010)	0.017* (0.010)
3 children	-0.029 (0.030)	-0.105** (0.045)	-0.030 (0.031)	-0.019 (0.030)
Educ, high school	-0.088*** (0.018)	-0.067*** (0.019)	-0.002 (0.013)	-0.006 (0.013)
Educ, vocational	-0.117*** (0.015)	-0.066*** (0.017)	-0.000 (0.011)	-0.036*** (0.011)
Educ, middle	-0.172*** (0.015)	-0.167*** (0.017)	-0.024** (0.011)	-0.045*** (0.011)
Educ, college	-0.160*** (0.015)	-0.099*** (0.016)	-0.023** (0.011)	-0.042*** (0.011)
Age -25	0.007 (0.014)	0.041*** (0.016)	-0.002 (0.010)	0.007 (0.011)
Age 31-35	0.007 (0.009)	0.004 (0.012)	-0.000 (0.007)	-0.000 (0.007)
Age 36-40	0.031*** (0.010)	0.007 (0.013)	0.001 (0.008)	0.006 (0.007)
Partner	-0.049*** (0.009)	-0.034*** (0.011)	-0.002 (0.007)	0.014** (0.007)
Income -200	0.143*** (0.011)	0.058*** (0.013)	0.005 (0.008)	0.038*** (0.008)
Income, 200-300	0.029*** (0.010)	-0.009 (0.014)	0.002 (0.009)	0.013 (0.009)
Income, 400-500	-0.018* (0.009)	0.029** (0.015)	0.011 (0.008)	0.003 (0.008)
Income, 500-	-0.005 (0.013)	0.070*** (0.018)	0.014 (0.010)	0.000 (0.009)
Constant	-0.688*** (0.017)	-0.322*** (0.019)	-0.021* (0.012)	0.013 (0.012)
N	6,671	6,628	6,652	6,621

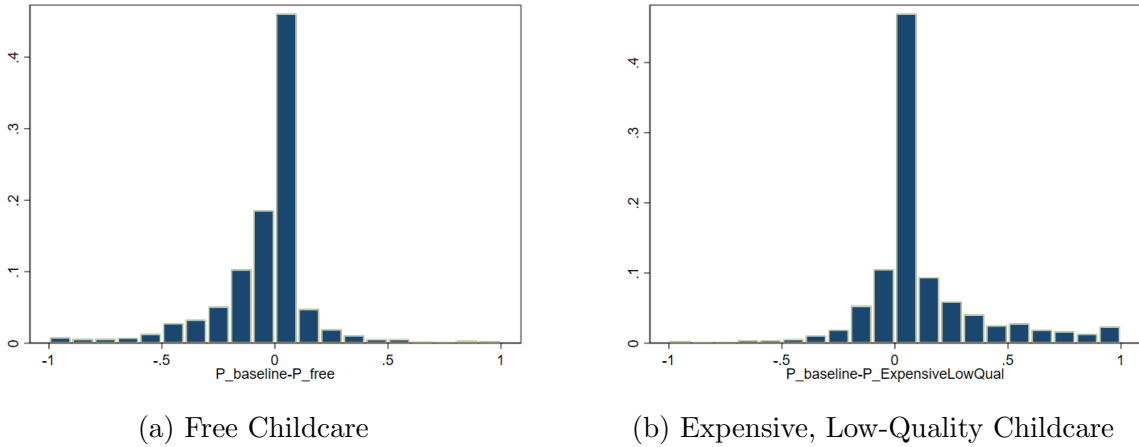
Notes: The table shows regressions of the $\widehat{seaITE}_s^i$ of Section 3.2.1, for $s = 3, 9, 18$, and 36 months after childbirth, on covariates measured in the administrative registries in 2018. Robust standard errors in parentheses. * 10%, ** 5%, *** 1% level of significance.

H Changes in Reported Fertility Probabilities across Policy Scenarios

Figure 6a reports average expected fertility in each policy scenario. Figure A4 reports the distribution across respondents of the individual change, the baseline report minus the scenario report of the probability of having a child within four years, for respondents who answered both questions. In both scenarios the distribution has a mass point at zero: a large share of respondents report the same probability under the policy as at baseline. Under free childcare the remaining mass lies mostly below zero, so most respondents who change their report raise their fertility probability, and a smaller share lower it. Under expensive, low-quality childcare the remaining mass lies mostly above zero, with a long right tail: some respondents who report a high probability at baseline report a probability near zero under the policy.

In the stochastic-choice formulation of de Paula and Tamer (2026), the identifying content of a policy or instrument state is the distribution of the changes it induces in individual treatment probabilities, and an instrument identifies an average of individual treatment effects weighted by those changes. Figure A4 is that distribution, measured respondent by respondent. Because the survey elicits each respondent's treatment probability under each regime, the counterfactual analysis in Section 3.5 uses these individual changes directly, through G_p^i regime by regime, rather than an average of them.

Figure A4: Change in Reported Fertility Probability, Baseline minus Scenario



Notes: Each panel is a histogram of $\hat{G}_b^i - \hat{G}_p^i$, the reported probability of having a child within the next four years in the baseline scenario minus the probability reported in policy scenario p , for respondents who answered both questions. Negative values indicate that the respondent reports a higher fertility probability under the policy than at baseline. Panel a is the free childcare scenario; panel b is the scenario in which the price of childcare is tripled and quality is lowered.